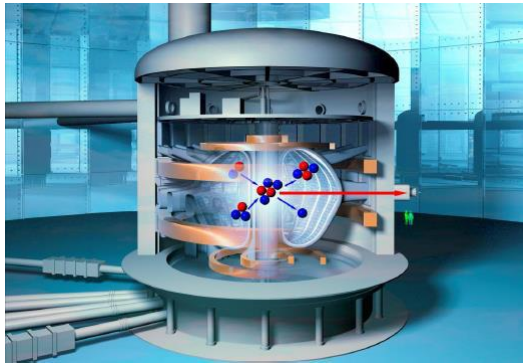




Coil Concepts for DEMO and Next Step Reactors

Peter H. Titus

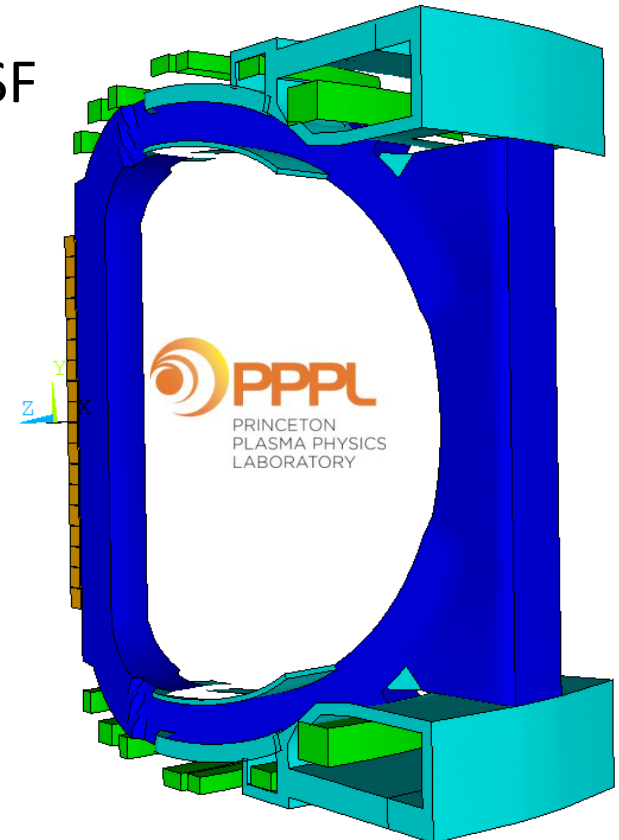
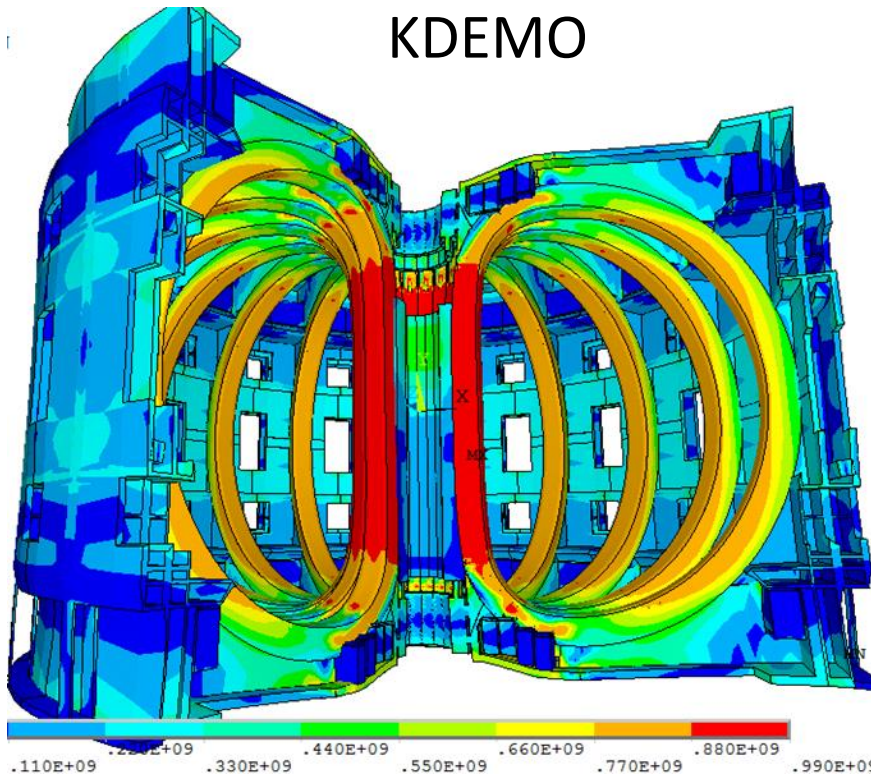
Princeton Plasma Physics Laboratory ptitus@pppl.gov, Analysis Group Head

5th IAEA DEMO Programme Workshop



KDEMO

US FNSF



Outline

Monotonic vs. Fatigue 3Sm, vs. Fracture vs Usual Monotonic Stress Checks

Limit Analysis for Determining Primary Stress

An Example KDEMO Evaluation

Non Constant Tension Dee Shapes

Never bending free, never filamentary

Thickness Variation to take OOP loads caused non-uniform tension stress

Designing for Out-of-Plane (OOP) loads with Radial and Vertical Maintenance

An Example US FNSF Evaluation

KDEMO as an Example of Vertical Maintenance

Remember to Size the Central Column based on a Full Scenario

An Example FNSF Evaluation

Older Aggressive Magnet Concepts - Useful for HTS magnets?

Use of Bucked and Wedged Concept for the US FNSF

DEMO/PILOT is Approaching Static Stress Criteria Determining Primary Stress Becomes More Important. Cyclic Life is Less Important Than Initial Fast Fracture

ITER 30,000 Shots

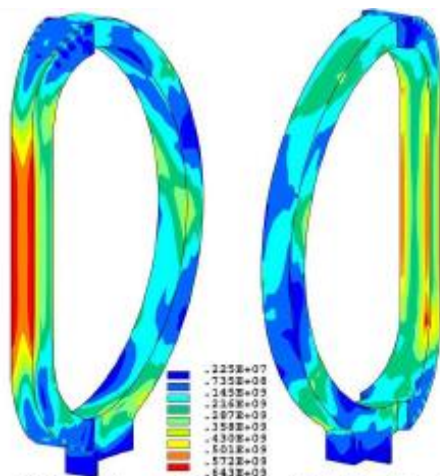


Fig. 3: Stress Intensity (Tresca) in TF coil casing

$$\frac{da}{dN} = C * (\Delta K)^m$$

Figure 6 shows a result of the LEFM approach.

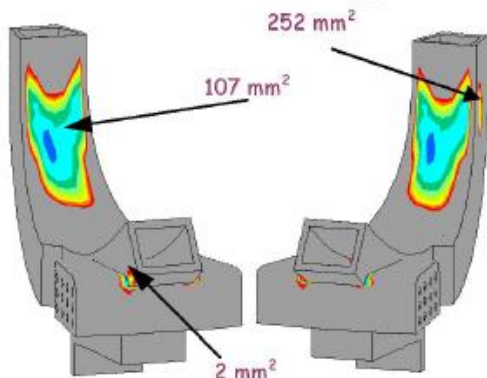
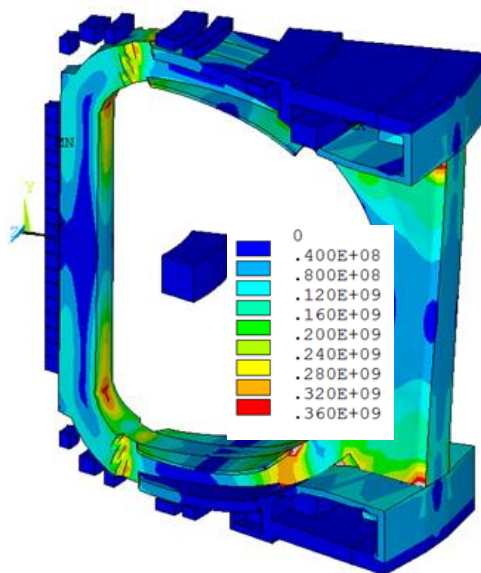


Fig. 6: Maximum allowable defect size lower part TF Coil casing

FNSF 1250 (?) Shots

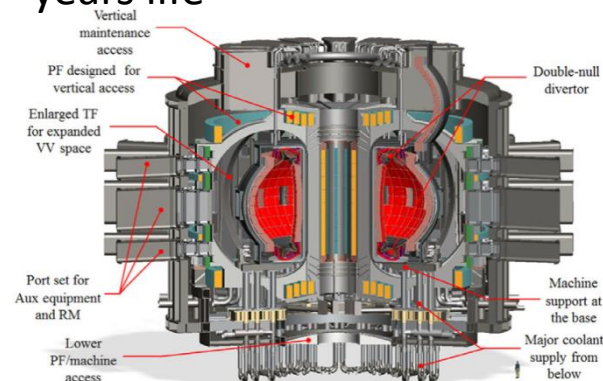
By it's nature to obtain significant fluences, pulses must be long . It has been estimated from the present FNSF program, approximately 700 DT shots, and probably another 500 He/H and DD shots would be experienced by the reactor in its design life.



Stress Difference between the TFON and the TFON+PF load case

KDEMO Two Phase Operation
Component Test Facility with
Multiple Pulses

Second Phase, Net Electricity
Generation 24/7 = 1 week
pulse =52 per year=1040 for 20
years life



DEMO/PILOT 20 to 50 Shots(?)
(No Experimental Phase)

**Much Less than Pulsed
Experimental Machines**

Monotonic Stress Checks

Satisfying S_m , $1.5 S_m$ and $3*S_m$ (or $2*S_m$)

Why is this important?

- **Tokamaks build out from the radius of the central column.**
- **There is a large cost impact of the radial build**
- **Small changes in the central column ripple through the rest of the radial build.**
- **Monotonic stress checks may limit the design, more so than fatigue.**

Primary Stress Allowable for Low Cycle – Non Fatigue Applications

Table 1. List of Materials for Magnet Metallic Components

Components	Material and Form	Minimum specified properties at 4K		
		Young's Modulus, Yield and Ultimate Strength	Fracture Toughness, K_{IC}	Paris Constants for Fatigue (R=0.1)
PF Coil Supports OIS Parts of Radial Plates and Covers	AISI 316 LN (General) as forged plates	205GPa $\sigma_y=800\text{MPa}$ $\sigma_u=1100\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=65.0\text{E-14m/cycle}$ $m=3.5$
CS Support Structure Top and Bottom Flanges	JK2 strengthened austenitic as forged plates	205GPa $\sigma_y=1000\text{MPa}$ $\sigma_u=1500\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=4.6\text{E-14m/cycle}$ $m=4.0$
TF Coil Case (inner leg regions), CS Pre-Load Structure Vertical Flanges, Gravity Supports	EK1 or JJ1 strengthened austenitic as forged sections	205GPa $\sigma_y=1000\text{MPa}$ $\sigma_u=1500\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=1.4\text{E-13m/cycle}$ $m=3.7$
CS Pre-Load Structure Vertical Flanges	Aluminium 7039	75GPa $\sigma_y=520\text{MPa}$ $\sigma_u=680\text{MPa}$		
TF Coil Case (outer leg regions)	EC1 strengthened austenitic as cast sections	190GPa $\sigma_y=750\text{MPa}$ $\sigma_u=1000\text{MPa}$	$K_{IC}=150\text{MPam}^{1/2}$	$C=5.77\text{E-14m/cycle}$ $m=3.73$
Parts of Radial Plates and Covers	AISI 316L as forged plates	205GPa $\sigma_y=700\text{MPa}$ $\sigma_u=1000\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=65.0\text{E-14m/cycle}$ $m=3.5$
IIS Poloidal Keys and OIS Bolts	Inconel 718	211GPa $\sigma_y=1350\text{MPa}$ $\sigma_u=1600\text{MPa}$	$K_{IC}=75\text{MPam}^{1/2}$	$C=8.26\text{E-15m/cycle}$ $m=4.55$
CS conductor jacket	Incoloy 908 as extruded square tubes, precipitation hardened	175GPa $\sigma_y=800\text{MPa}$ $\sigma_u=1100\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=28.4\text{E-14m/cycle}$ $m=3.58$
TF conductor jacket, CS conductor jacket	Modified and aged 316 LN for Nb3Sn, as extruded circular tube or square sections	205GPa $\sigma_y=1000\text{MPa}$ $\sigma_u=1600\text{MPa}$	$K_{IC}=150\text{MPam}^{1/2}$	$C=3.86\text{E-11m/cycle}$ $m=2.394$
PF conductor jacket	AISI 316L as extruded square tube	205GPa $\sigma_y=700\text{MPa}$ $\sigma_u=1000\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	$C=65.0\text{E-14m/cycle}$ $m=3.5$
Cooling pipes	AISI 316L as extruded (seamless) circular tube	205GPa $\sigma_y=700\text{MPa}$ $\sigma_u=1000\text{MPa}$	$K_{IC}=200\text{MPam}^{1/2}$	

TF Case and Winding Pack Analysis

TF Case

$2/3 * \text{yield} = 666 \text{ MPa}$

$1/2 \text{ Ultimate} = 750 \text{ Mpa}$

$1/3 \text{ Ultimate} = 500 \text{ MPa}$

S_m , Primary Membrane Allowable = 666 Mpa Only based on yield according to ITER MSDC

According to PPPL Criteria S_m = lesser of $2/3$ Yield and $1/2$ Ultimate + Show Adequate Ductility

The average stress in the inner leg case should satisfy this Allowable

From the PPPL NSTX (FIRE) Criteria (Not Really Intended for Superconducting Magnets)

For support structures and any other **STEEL** structures including, if applicable, the vacuum vessel, the design Tresca stress values (S_m) shall be based on the lesser of the following:

2/3 of the *minimum* specified yield at temperature

Reference 3 (Section III, Appendix III, Article III-2110(a)(3))

1/2 of the *minimum* specified tensile strength at temperature

Original 1/3 S_u per Ref. 3 (Section III, Appendix III, Article III-2110(a)(1)).

3* S_m (twice yield) is allowed

An over-simplification but ASME is based on a factor of safety of 2 to 3 against failure

From the ITER Magnet Criteria

Yielding at cryogenic temperatures is accompanied by substantial temperature changes in the material (several tens of degrees) which can produce the well-known phenomenon of serrated yielding in test specimens. The measurement problems associated with material behaviour above the yield stress again mean that a design based on the yield stress is preferred.

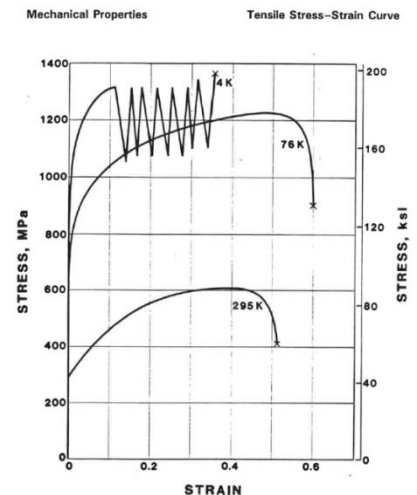


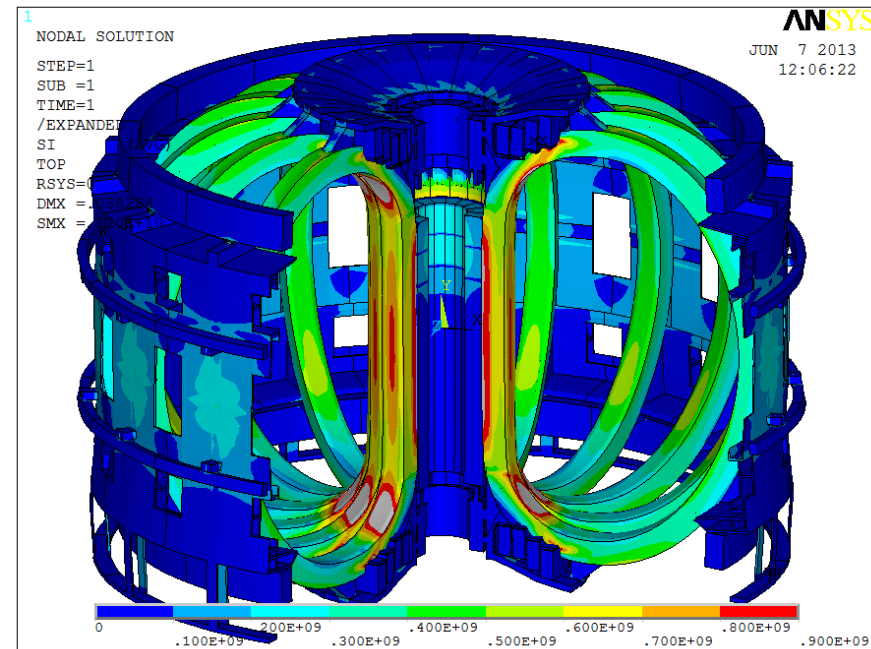
Figure 4. Engineering stress versus strain curves at three temperatures for annealed (1330 K, 1 h) 316LN of the following composition in weight percent: 17.4Cr, 13.9Ni, 2.53Mn, 1.58Mo, 0.48Si, 0.16N, 0.016C, 0.021P, S. The specimens were 6-mm-diameter bar, and strain was measured with a strain-gage extensometer of 25-mm gage length. Source: Read and Reed (1978).

Primary Stress

Primary stress is any normal stress or a shear stress developed by an imposed loading which is necessary to satisfy the laws of equilibrium of external and internal forces and moments. The basic characteristic of a primary stress is that it is not self-limiting. Primary stresses which considerably exceed the yield strength will result in failure or, at least, in gross distortion. A thermal stress is not classified as a primary stress.

From the ITER Criteria:

The relatively low value of σ_u / σ_y at cryogenic temperatures (about 1.5 at 4K compared to 2.5 at 200C) means that a design based on the ASME definition of S_m would be dominated by tensile strength. For ITER, therefore only the yield stress is used to define S_m . The function of the ITER static stress limit is to limit the onset of plastic flow in the material, for which σ_y is the appropriate measurement to use as a base. Consistent with the ASME philosophy for safety factors, **the peak stresses in ITER are limited absolutely to $2.0S_m$ (i.e. below the ultimate stress, which is more conservative than ASME)** and in some cases (generally where local plasticity may affect insulation bonding) to $1.5S_m$.

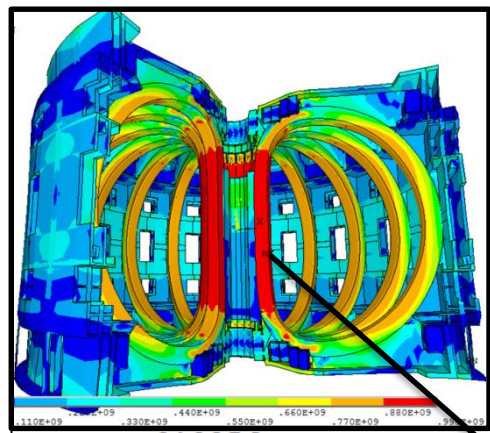
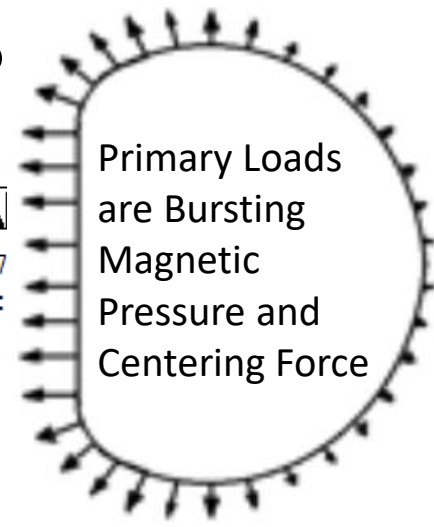


For a DEMO reactor like KDEM0, what is the primary stress , Where is plasticity allowed? How much?

What is the Primary Stress in the TF Inner Leg?

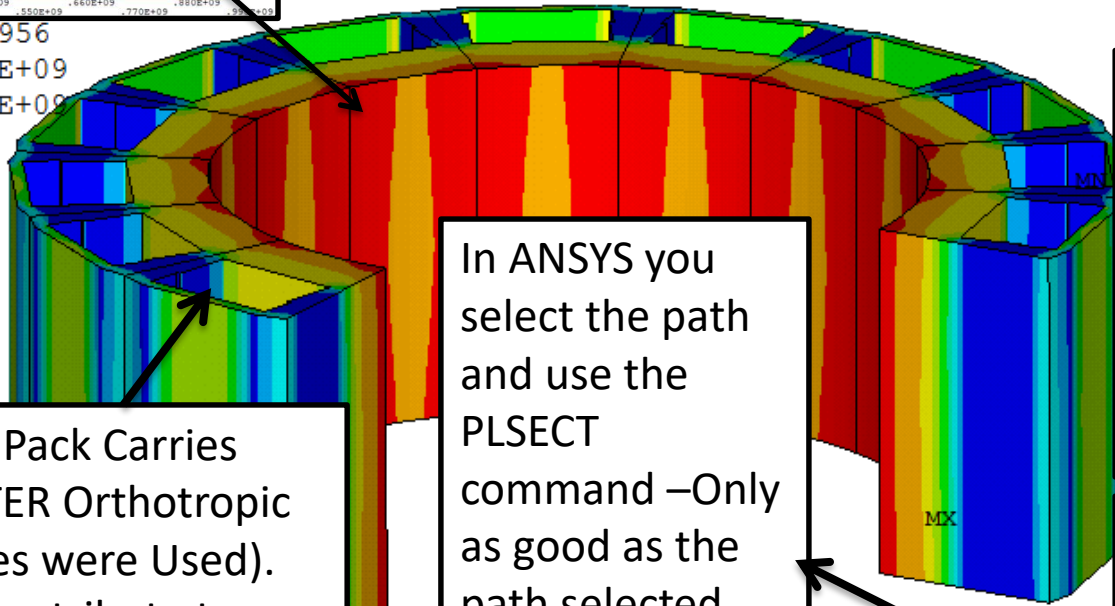
Especially based on FEA.

What is the contribution of The Winding Pack?



This is a KDEMO Analysis, 7 T at Ro=7m

DMX = .010956
SMN = .612E+09
SMX = .930E+09



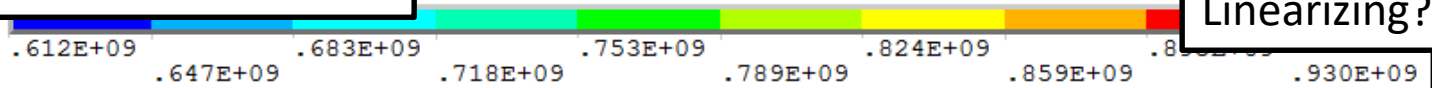
Winding Pack Carries Loads (ITER Orthotropic Properties were Used). Does it Contribute to Carrying the Primary Load?

In ANSYS you select the path and use the PLSECT command –Only as good as the path selected

TF Case
2/3* yield = 666 MPa
1/2 Ultimate = 750 MPa
1/3 Ultimate = 500 MPa

Allowable = 666 MPa
According to ITER MSDC

“Average” = 859 MPa
By “eyeball”
Question: What Section Should be used for Linearizing?



Limit Analysis

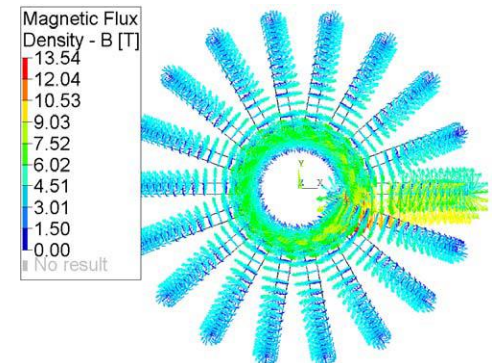
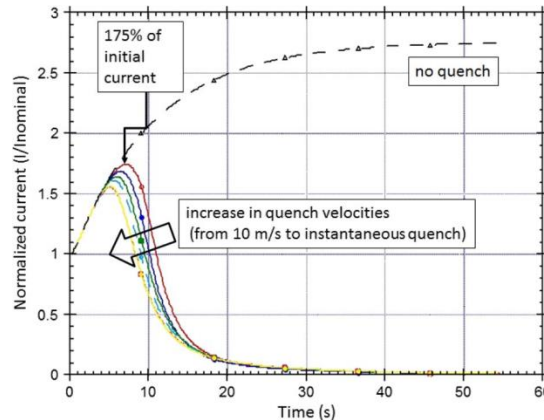
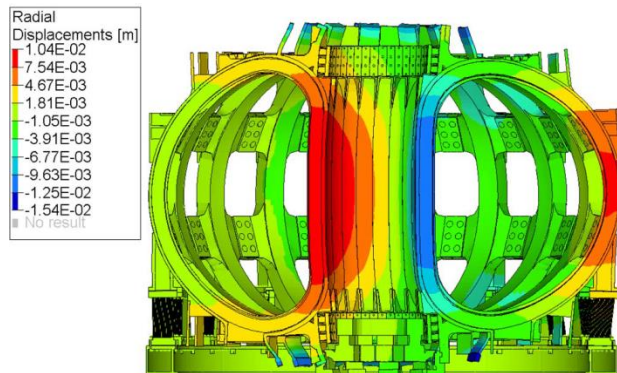
From the NSTX structural criteria

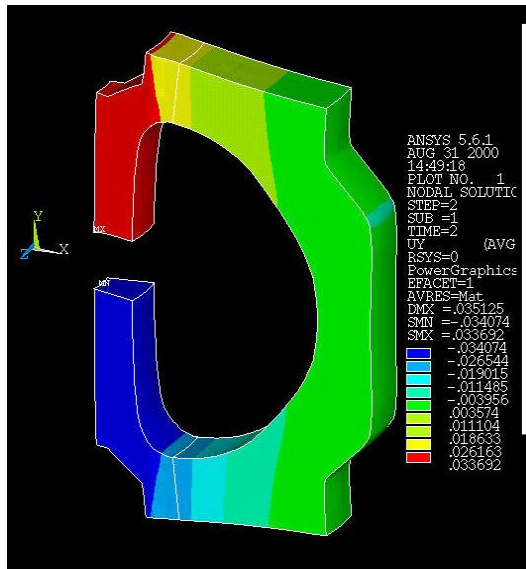
"An exception to this elastic analysis approach can be when the nature of the structure and its loading make it difficult to decompose the stresses into the above mentioned categories. In such an instance, a detailed, non-linear analysis that accounts for elastic-plastic behavior, frictional sliding and large displacement shall be used to determine the limit load on the structure. The limit load is that load which represents the onset of a failure to satisfy the Normal operating condition as described in Section I-2.6. The safety factor of limit load divided by the normal load shall be greater than 2.0."

Similar wording was in the ITER Magnet Structural Design Criteria (MSDC) The objection was that it was non-physical and the coils would quench before reaching the load factor of 2.0.

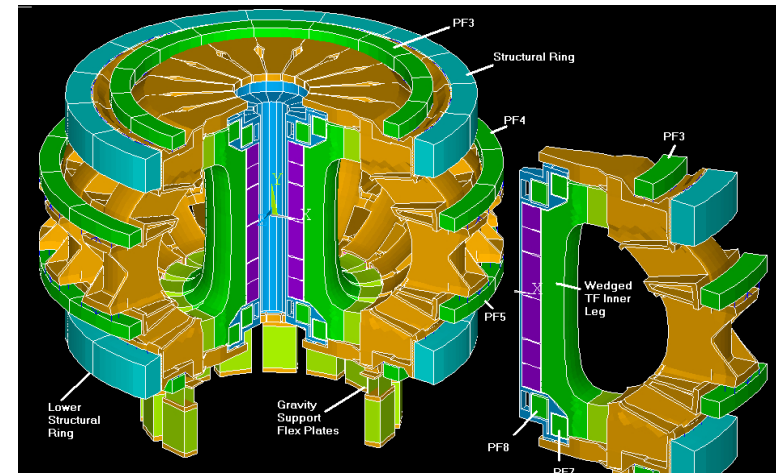
However this is only intended as a "thought experiment", only to demonstrate structural factors of safety, and replace linearization and judgmental determinations of primary, bending discontinuity and peak stresses.

Currents can temporarily go beyond quench initiation – in Fault cases. Below are some results for an ITER shorted TF coil evaluation

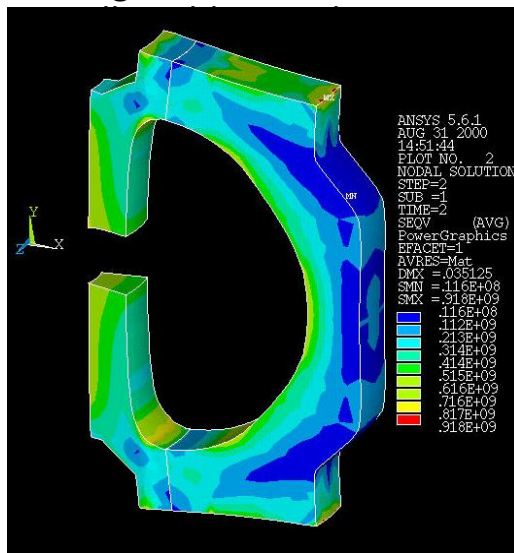




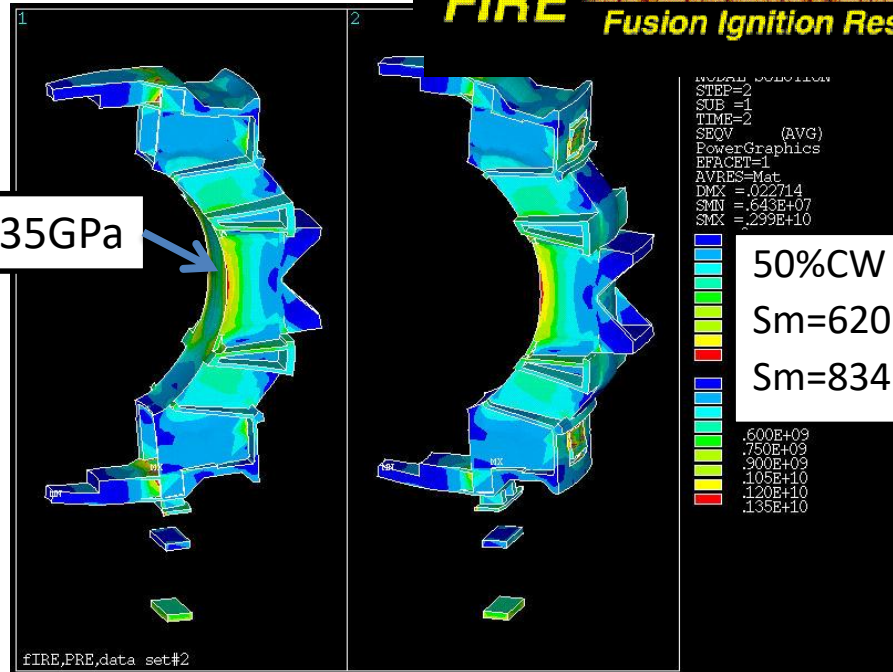
Can We Argue that
 the Inner Leg Stress
 is not the Primary
 Stress? What If the
 Inner Leg Cannot Take
 Vertical Tension? An
 Extreme Example Taken
 from FIRE



- Separation is 6 cm,
 Average stresses in outer
 leg are at membrane



1.35GPa



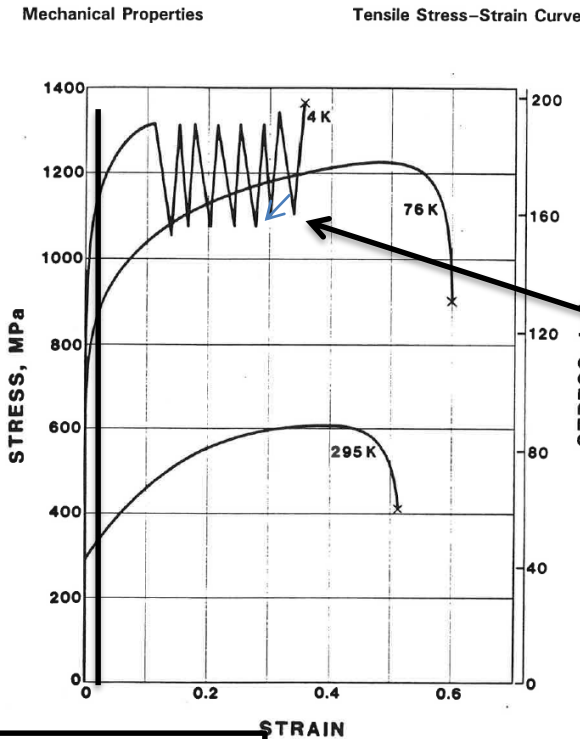
50%CW 304 SST
 Sm=620MPa at RT
 Sm=834MPa at 80K

The argument is not completely successful, but it illustrates
 that the outer structures take a larger share of the Primary
 bursting load

KDEMO Elastic-Plastic Limit Analysis.
Must Show a factor of 2 on failure (Non-Convergence in ANSYS Model)

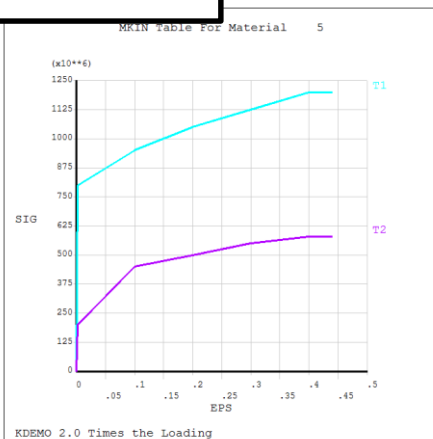
Results of KDEMO Limit Analysis

The plasticity needed to demonstrate a load factor of 2 is below the level of yield serrations – at least in this measured stress strain curve.



KDEMO Strain
at Load Factor
of 2.0

Strain curves at three temperatures for annealed composition in weight percent:
4N, 0.48Si, 0.16N, 0.016C, 0.021P, S.
and strain was measured with a strain-gage
force: Read and Reed (1978).



ANSYS 14.0.3
JUN 25 2013
15:45:46
Table Data
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T2= 292.00
ZV =1
DIST=.75
XF =.5
YF =.5
ZF =.5
Z-BUFFER

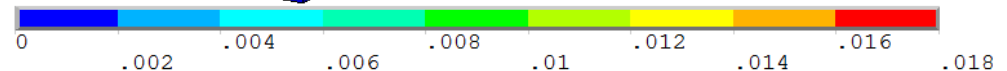
1
NODAL SOLUTION
STEP=7
SUB =3
TIME=7
/EXPANDED
EPPLEQV (AVG)
TOP
RSYS=0
DMX =.071524
SMX =.060456

ANSYS

JUN 25 2013
22:44:32

.01(1%) Strain

Plastic Strain at Twice Normal Loading



KDEMO 2.0 Times the Loading

1

ANSYS

JUN 25 2013

22:40:52

NODAL SOLUTION

STEP=8

SUB =6

TIME=8

/EXPANDED

UY

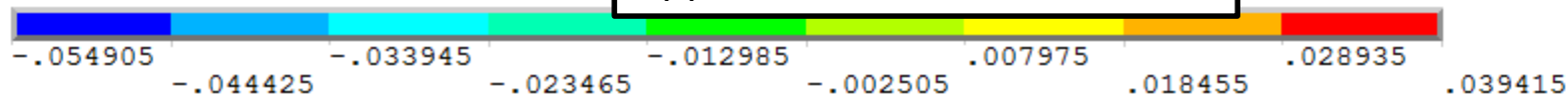
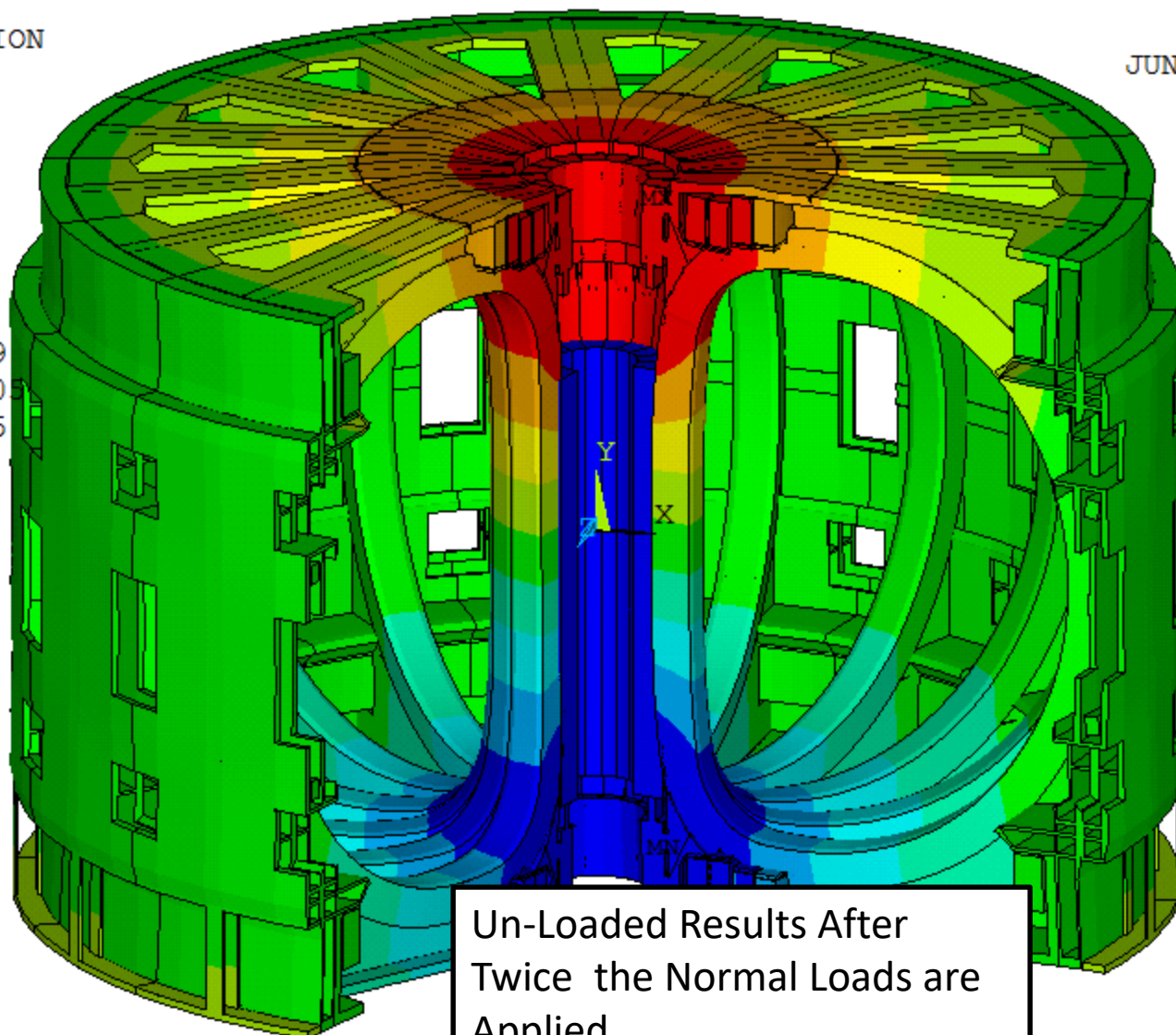
TOP

RSYS=0

DMX =.055489

SMN =-.054905

SMX =.039415



KDEMO UnLoaded

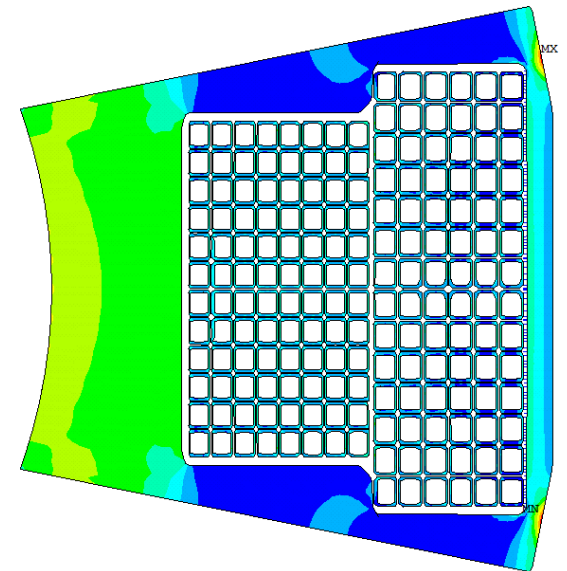
In the analysis presented, failure was defined as a plastic collapse indicated by a failure of the elastic plastic analysis to converge. It is a “thought experiment” intended to quantify structural factor of safety, but it could be extended to other failure mechanisms to establish their factors of safety

Failure could be defined as:

Exceeding an Insulation Strength or Strain Limit

Exceeding the irreversible Nb₃Sn or HTs strain limit

Stress Intensities exceeding the fracture toughness



The limit analysis model would have to be expanded to include details of the winding pack

For Low Cycle Qualification, Use preservice ASME and API qualifications based on Non-Destructive Examination of cracks . Perform fast fracture assessment based on the maximum undetected flaw size.

Example of ANSYS
KCALC -
Quantifies stress
intensity for
identified initial
crack

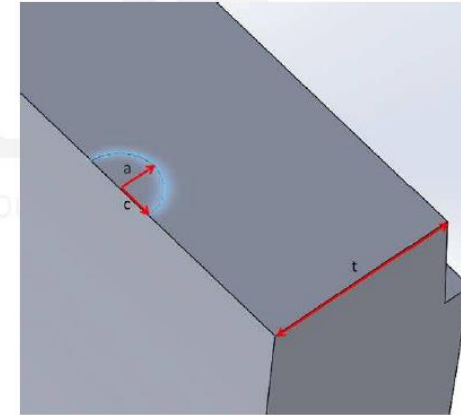
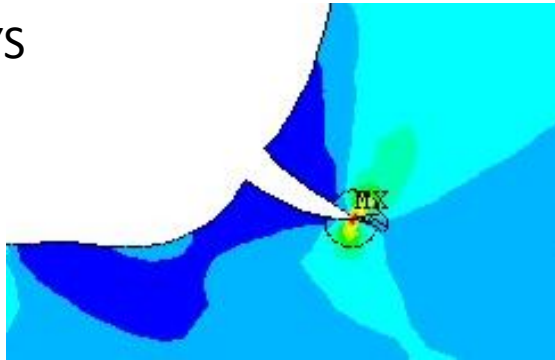
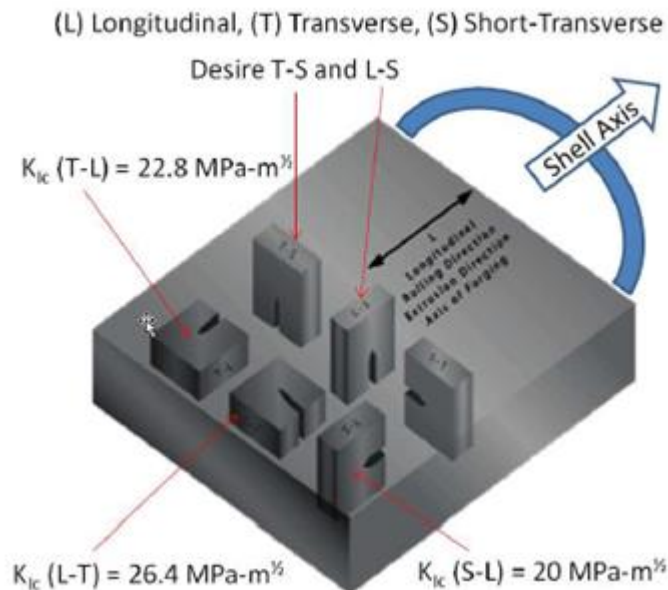


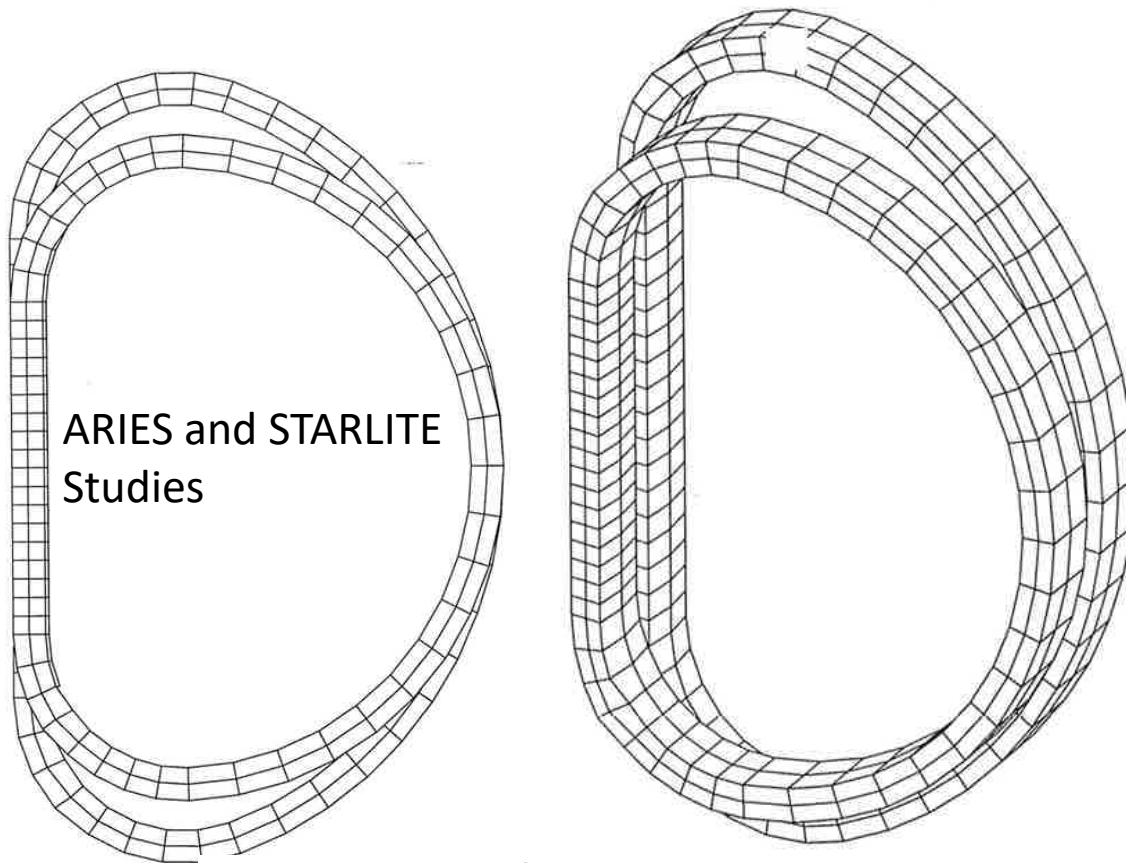
Figure 5 Radial Thumbnail Crack geometry shown at Mid-Plane of shell



ITER requires 2 on crack size, 2 on Life, and 1.5 on Fracture toughness

In the limit, with no crack propagation, and 2 on crack size intended for NDE uncertainty, This would leave a Factor of Safety against burst of 1.5 – **Consider Fracture Toughness F.S. of 2.0 for DEMO**

Next Topic: Deviation From Constant Tension D



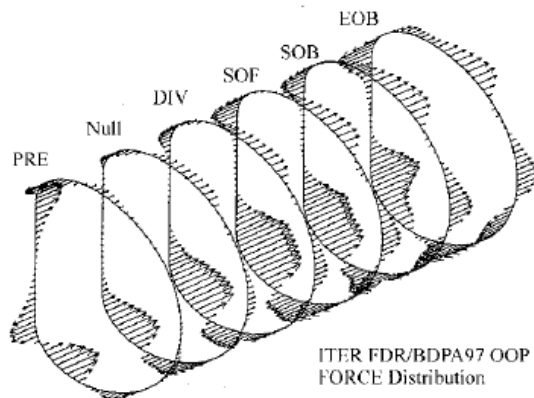
In early ARIES studies, particularly ARIES RS, “squashing” the constant tension Dee form was investigated.

There was a penalty in extra bending stresses, but

There was a benefit in having the PF coils more effective at the plasma, and

This allowed reducing the PF currents, and reducing the out-of-plane loads which were a more significant source of bending.

-More on this relating to the US FNSF



ARIES-RS magnet systems

L. Bromberg ^{a,*}, P. Titus ^b, J.S. Schultz ^a, M. Sidorov ^a, S. Pourrahimi ^a,
The ARIES Team

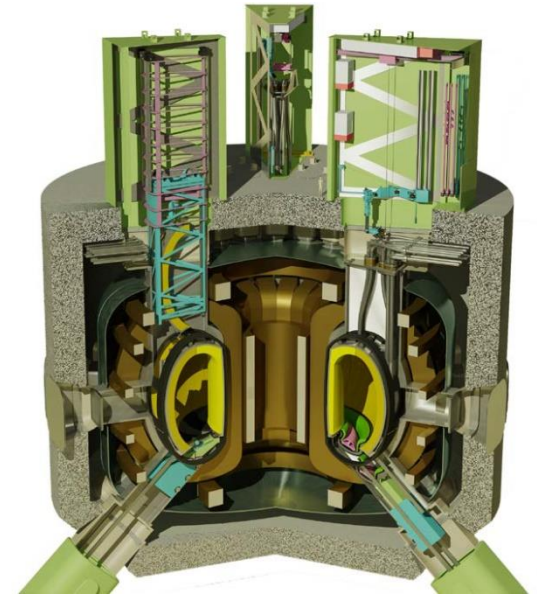
^a Plasma Fusion Center, MIT, Cambridge, MA 02139, USA

^b Stone and Webster, Boston, MA 02215, USA

Next Topic: Radial Servicing vs. Vertical Servicing

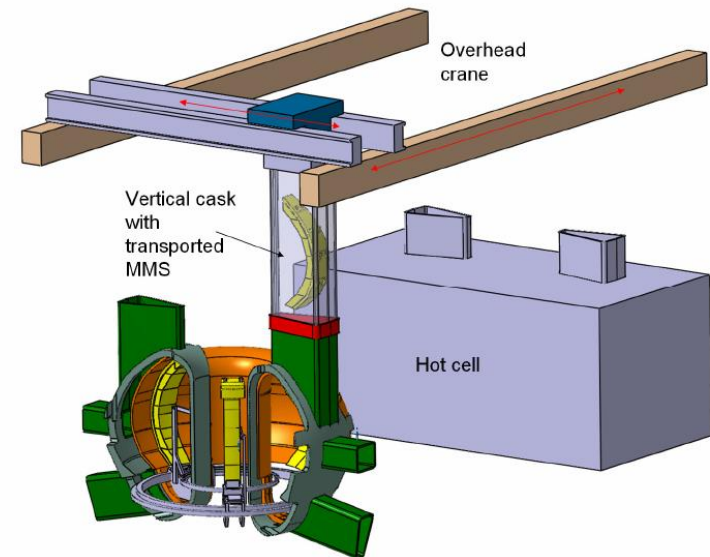
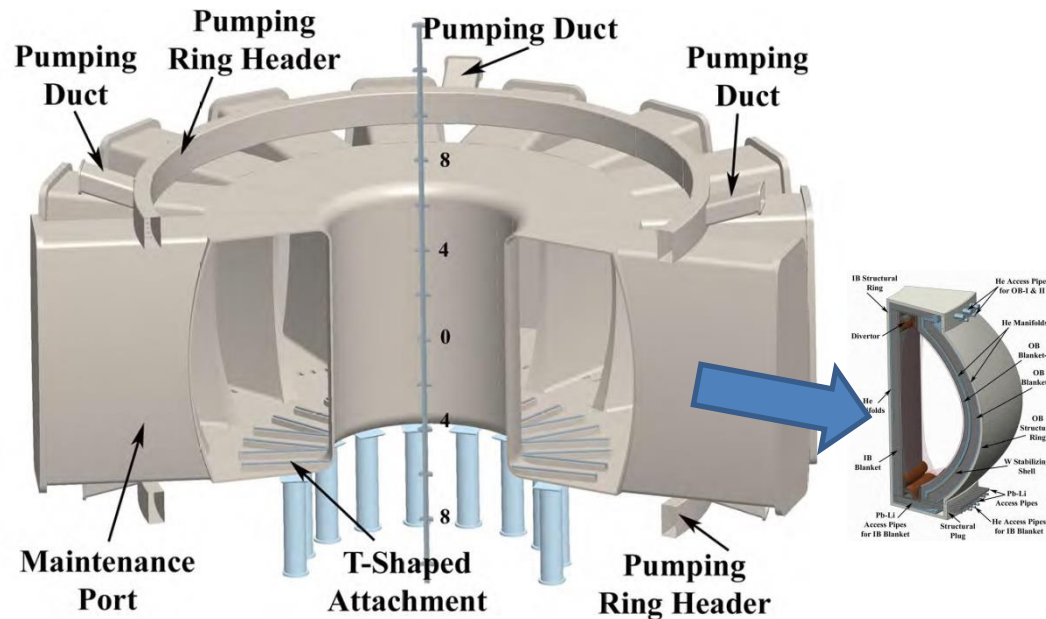
The US FNSF servicing logic relies on radial servicing which requires a large opening between the outer legs to extract the core components.

In most other next generation machines, this space is taken up by torque structures and external components like Neutral Beams. Many competing next generation machines use vertical servicing through openings between the horizontal legs of the TF coils. KDEMO and EU Demo are examples.

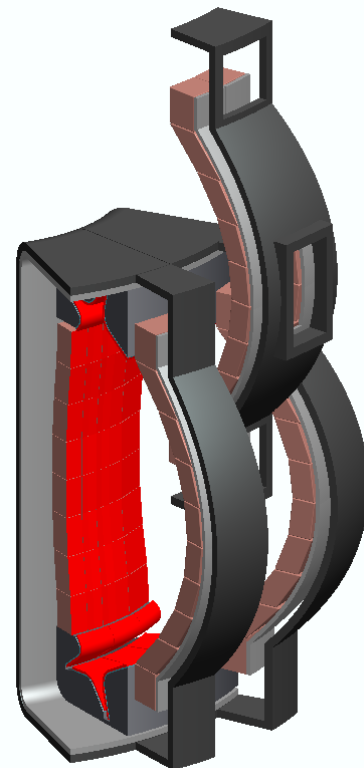
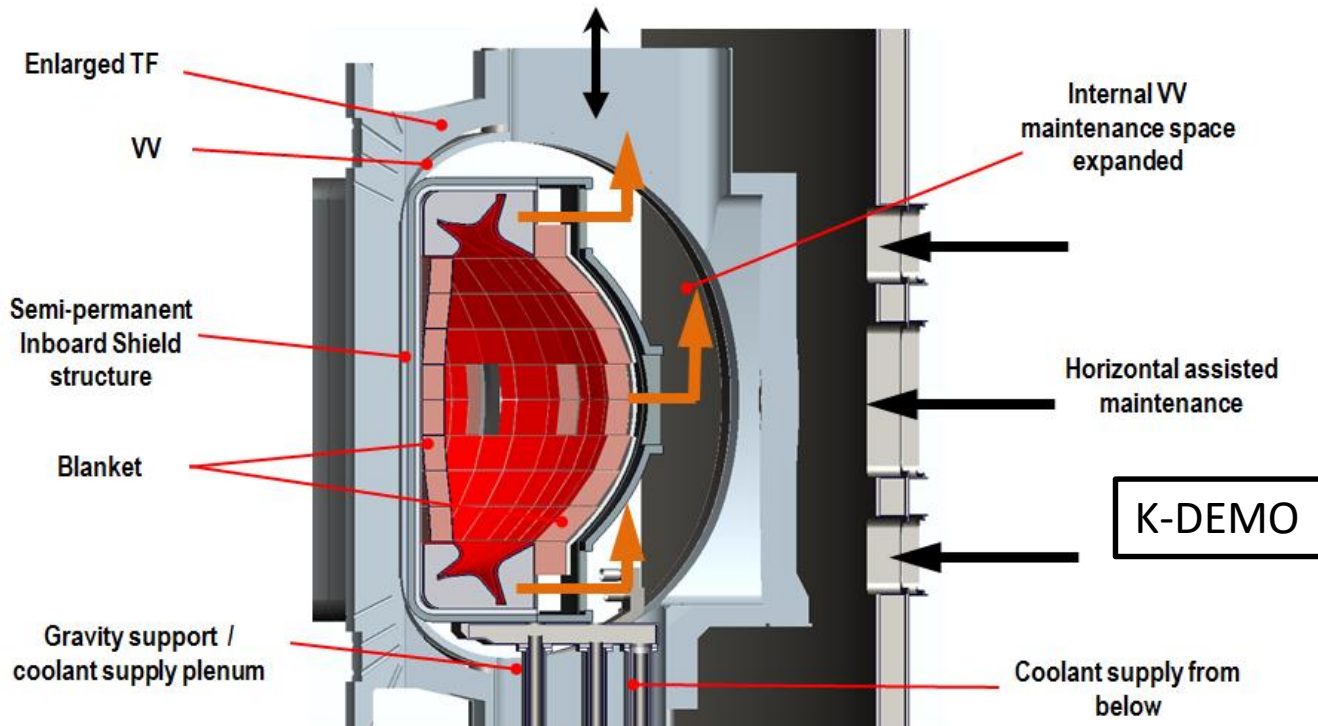


EU Demo with Vertical Servicing

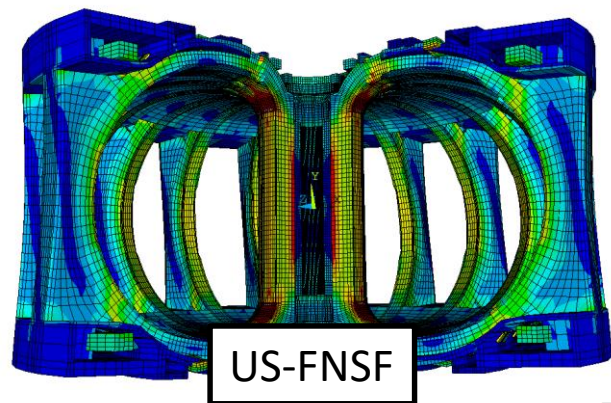
Aries ACT and the US FNSF Study



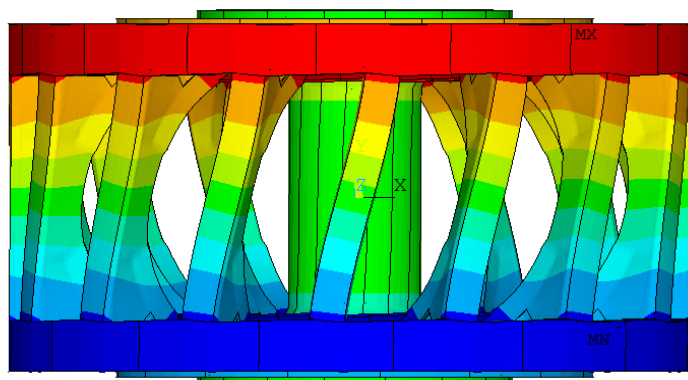
Vertical Servicing Requires a Large Open Span on Top



Radial Servicing Requires a Large Opening on the Machine Outer Radius



US-FNSF

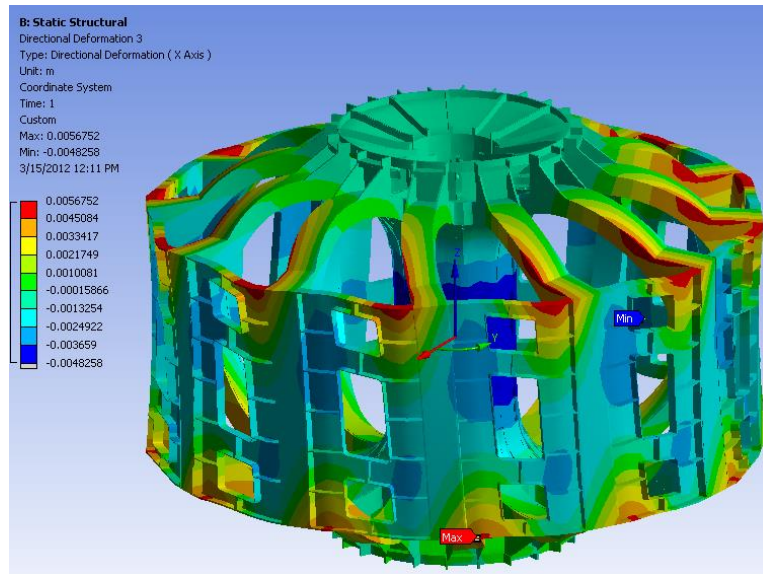


FNSF Current Set #2

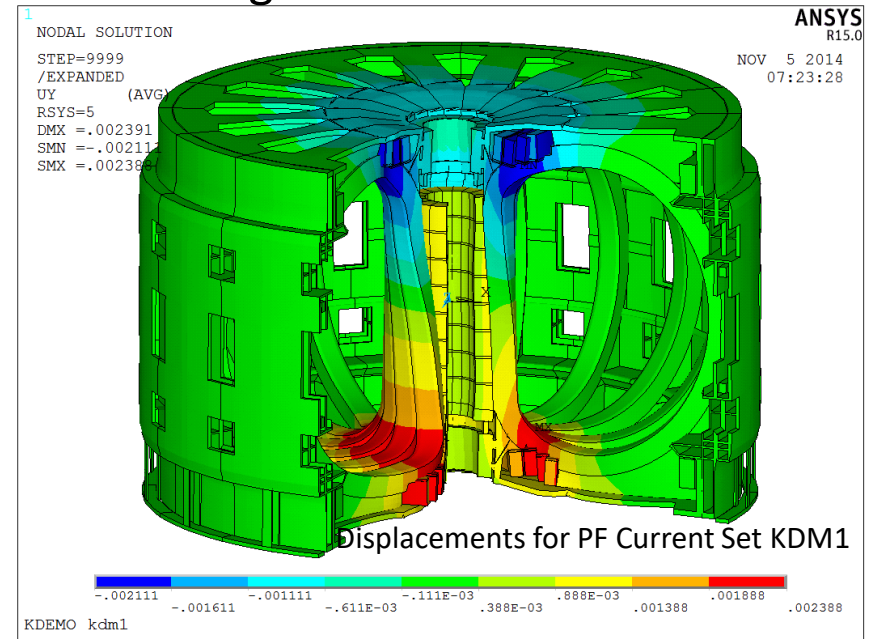
STEP=9999
/EXPANDED
UY (A
RSYS=5
PowerGraphi
EFACET=1
AVRES=Mat
DMX =.01341
SMN =-.0134
SMX =.01341
-.013
-.010
-.007
-.004
-.001
.0014
.0044
.0074
.0104
.0134

In the case of the FNSF, the vertical span of the outer leg must be stiff and strong enough to carry the global torque as beam elements connecting upper and lower structures.

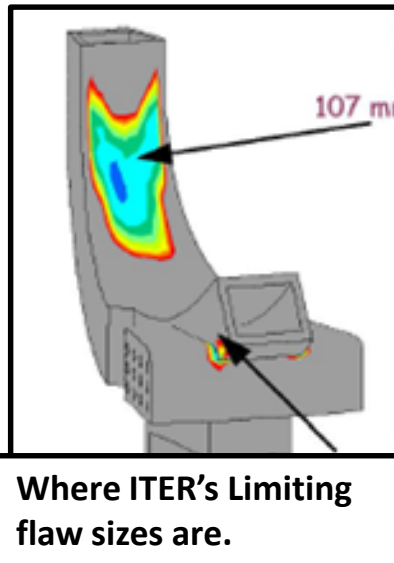
Vertical Maintenance is not Without Structural Challenges – The “Window” is Simply on Top Rather than the Side



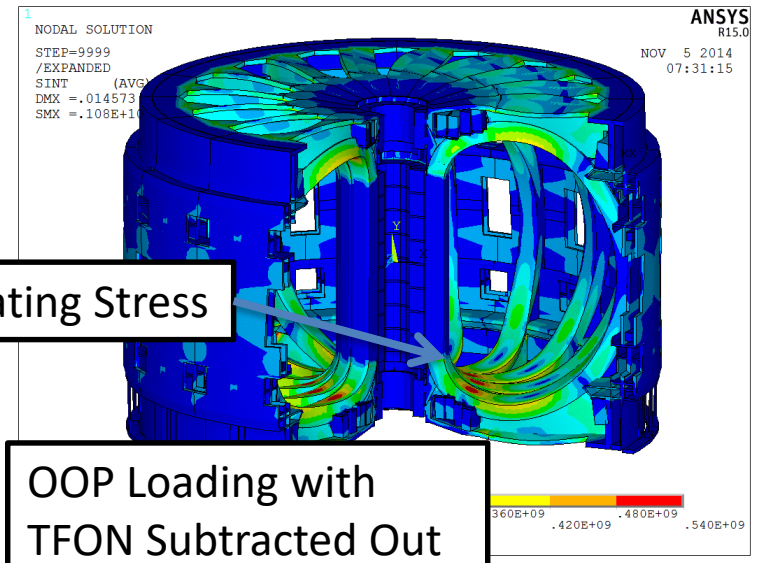
KDEMO Before Outer Structure Reinforcements - +/- 5mm



KDEMO Toroidal Displacement OOP Loading with TFON Subtracted Out +/- 2mm With Top and Outer Reinforcement



540 Mpa Alternating Stress



1

ANSYS

JUN 7 2013

12:06:22

NODAL SOLUTION

STEP=1

SUB =1

TIME=1

/EXPANDED

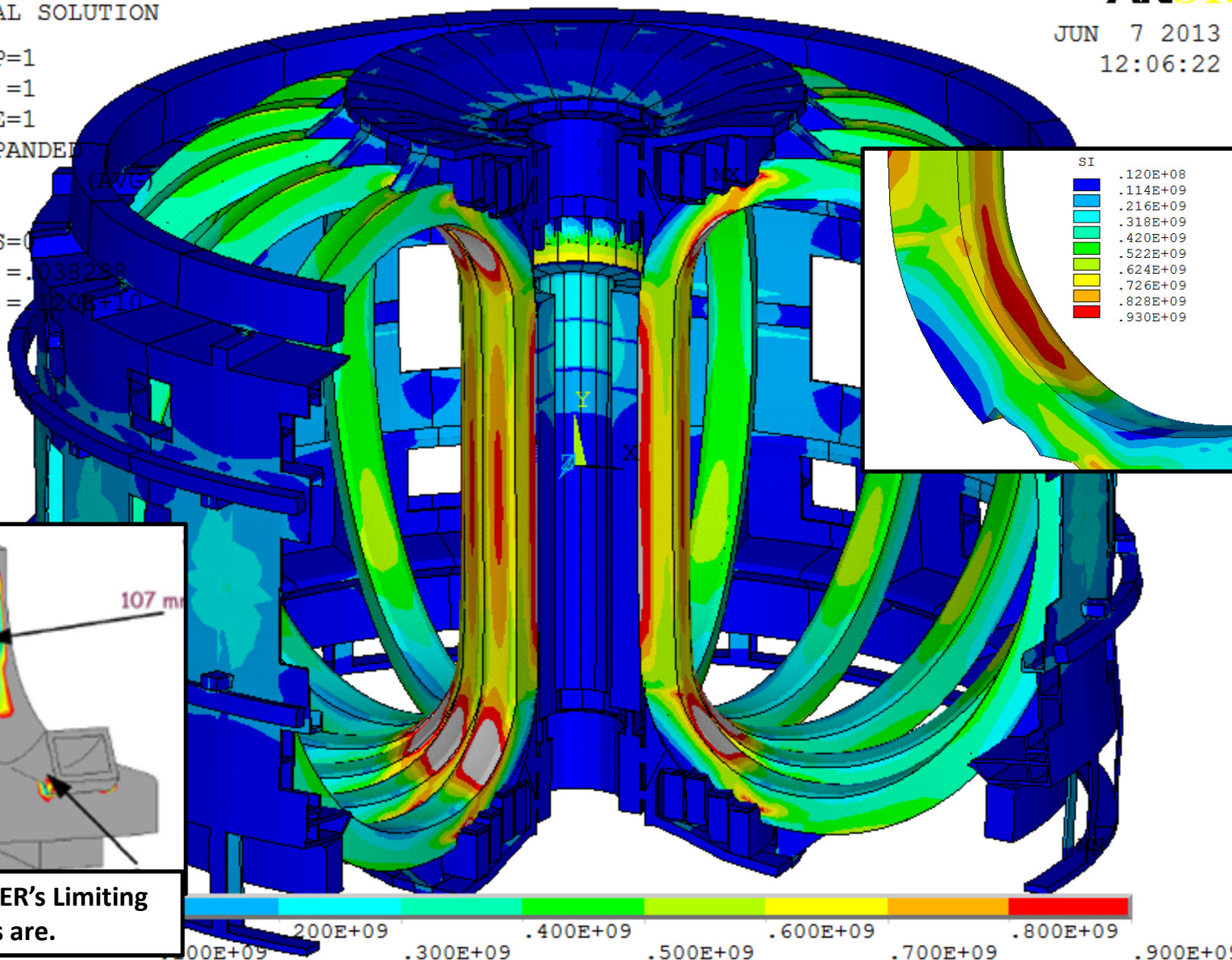
SI

TOP

RSYS=0

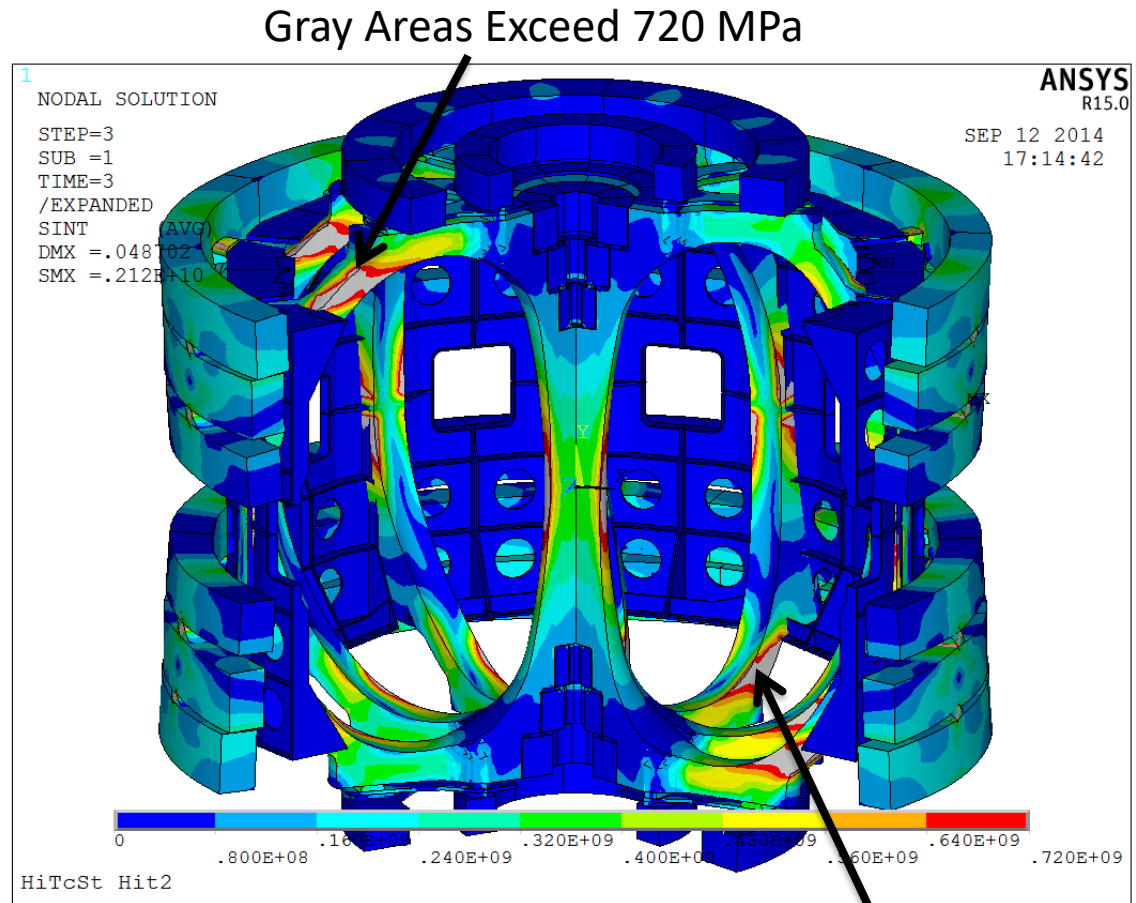
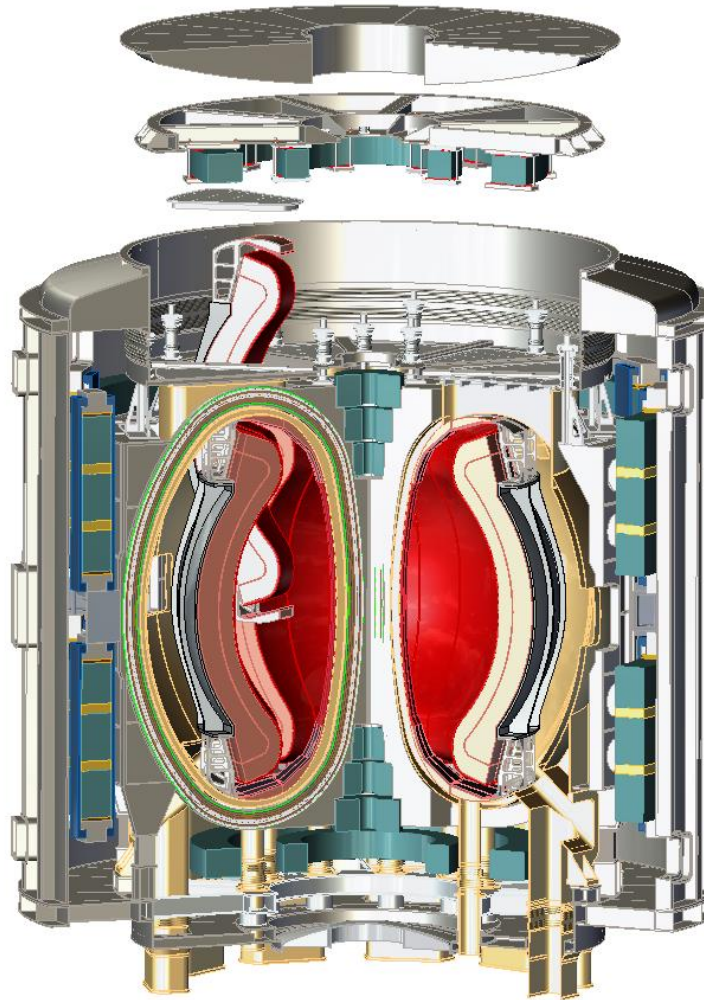
DMX = .038288

SMX = .120E+10



HTS ST Vertical

**Maintenance Concept Studied at PPPL (NSTXU is a ST)
also had TF Bending Issues at Vertical Port**

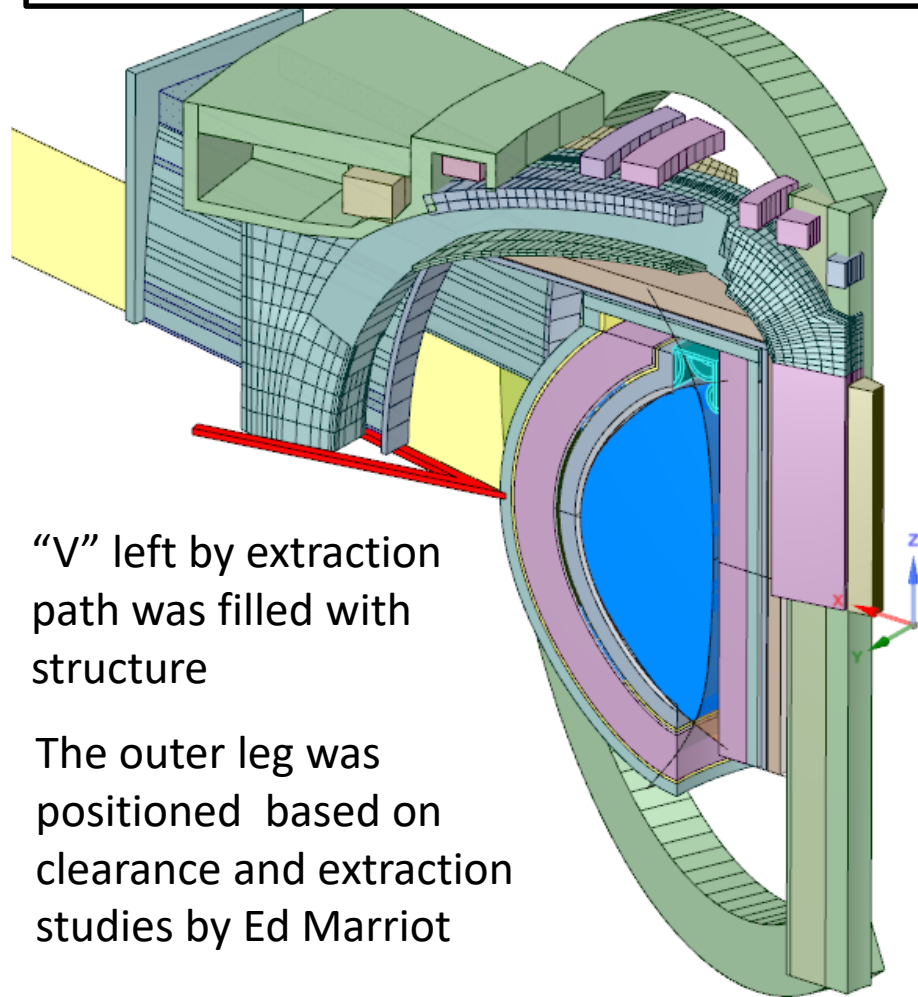


A separate cryostat allows the TF and remaining PF coils to retain their cryogenic state during removal of plasma components.

Back to the US FNSF :

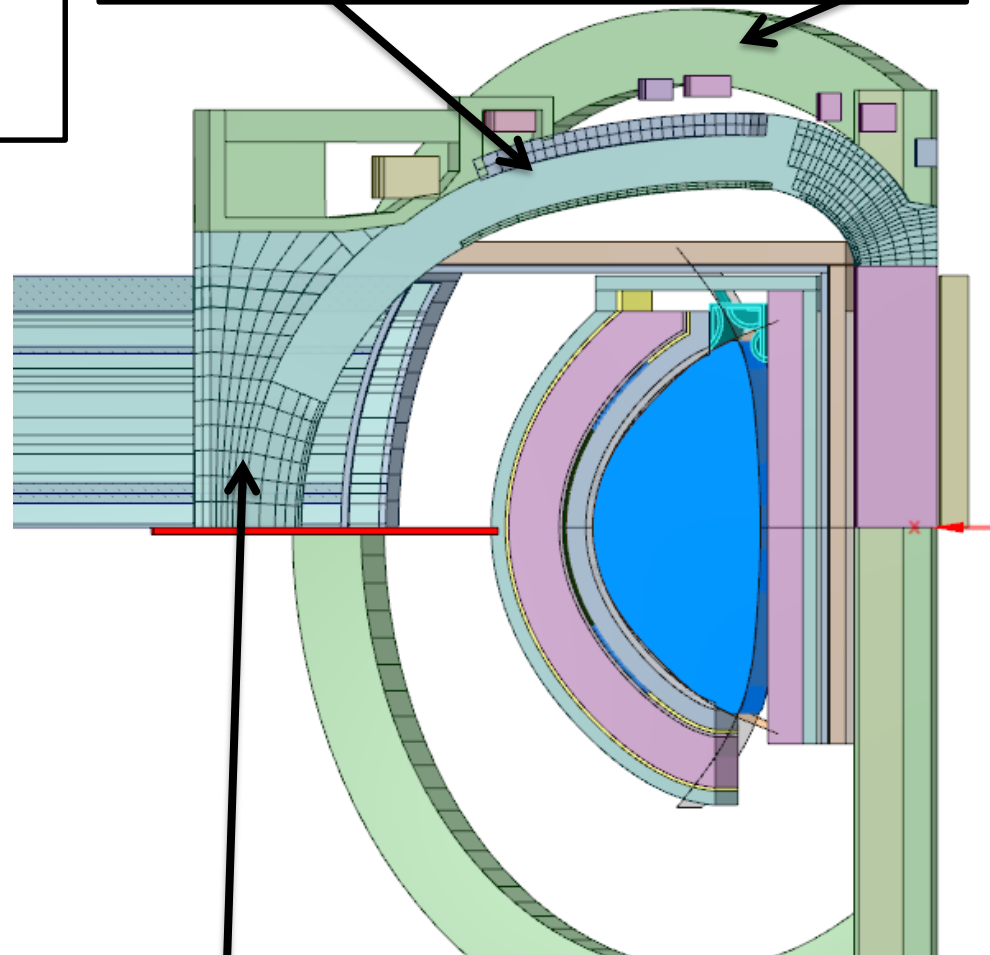
“Squashed” Shape Allows PF coils to be closer to the plasma and allows the outer leg to be moved out radially

Shape closer to Constant Tension “D” from the Systems Code.
“Squashed” shape used for the FNSF.



“V” left by extraction path was filled with structure

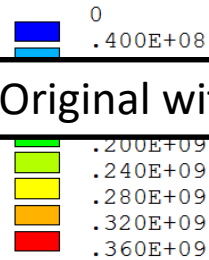
The outer leg was positioned based on clearance and extraction studies by Ed Marriot



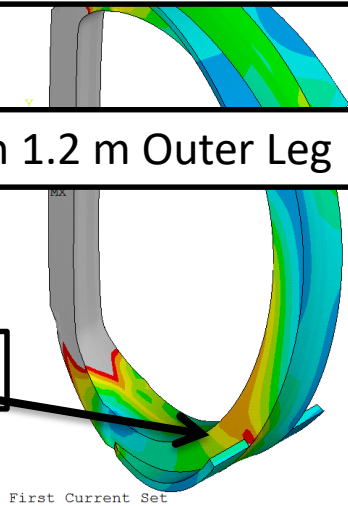
The FEA mesh was converted to IGES solids and sent to Ed Marriott for insertion into the CAD model for the final clearance checks

The Analysis Evolved - We kept adding outer structure

Original with 1.2 m Outer Leg

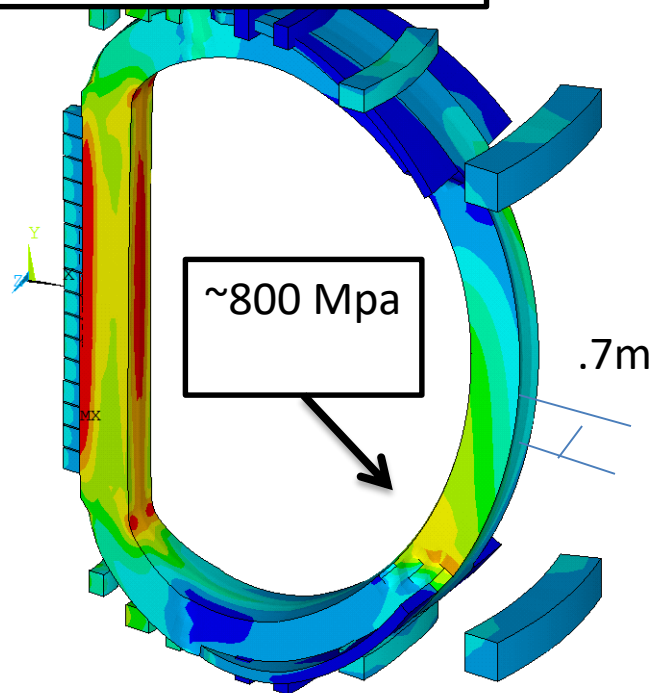


~369 Mpa



In the original Aug 2015 analysis, the outer leg dimension was a guess at 1.2 m. In the next analyses I thinned the outer leg and added structures radially outward and above and below the service port

Thinned to .7 m Outer Leg



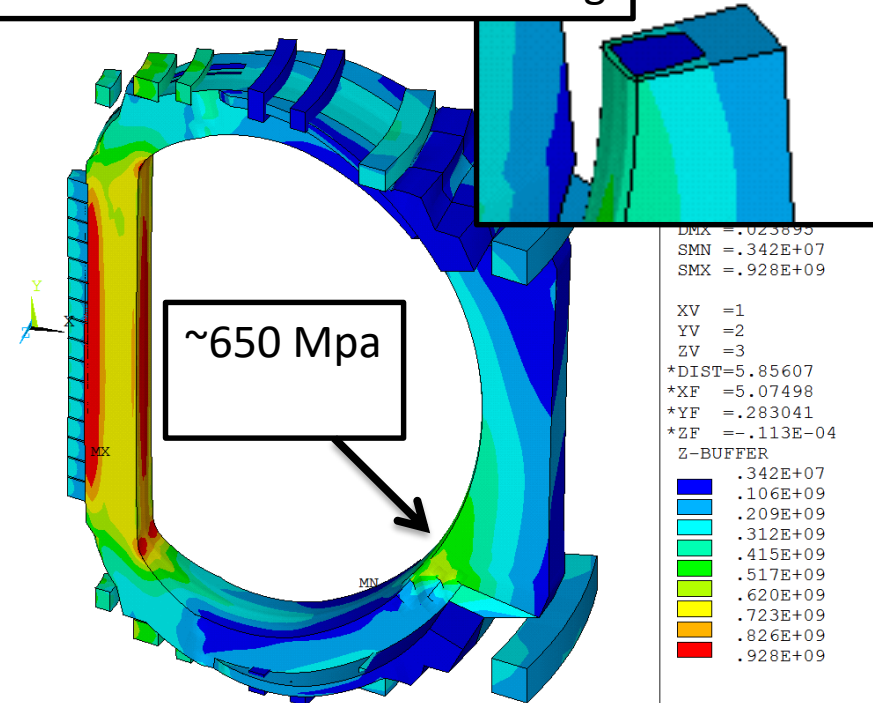
~800 Mpa

ANSYS
R15.0

AUG 27 2015
 15:00:42
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 STEP=1
 SUB =1
 TIME=1
 SINT (AVG)
 PowerGraphics
 EFACET=1
 AVRES=Mat
 DMX =.023895
 SMN =.849E+07
 SMX =.930E+09

XV =1
 YV =1
 ZV =2
 *DIST=6.14728
 *XF =5.25435
 *YF =.246677
 *ZF =-.113E-04
 Z-BUFFER
 .849E+07
 .111E+09
 .213E+09
 .316E+09
 .418E+09
 .520E+09
 .623E+09
 .725E+09
 .828E+09
 .930E+09

Thinned and Extended Outer Leg

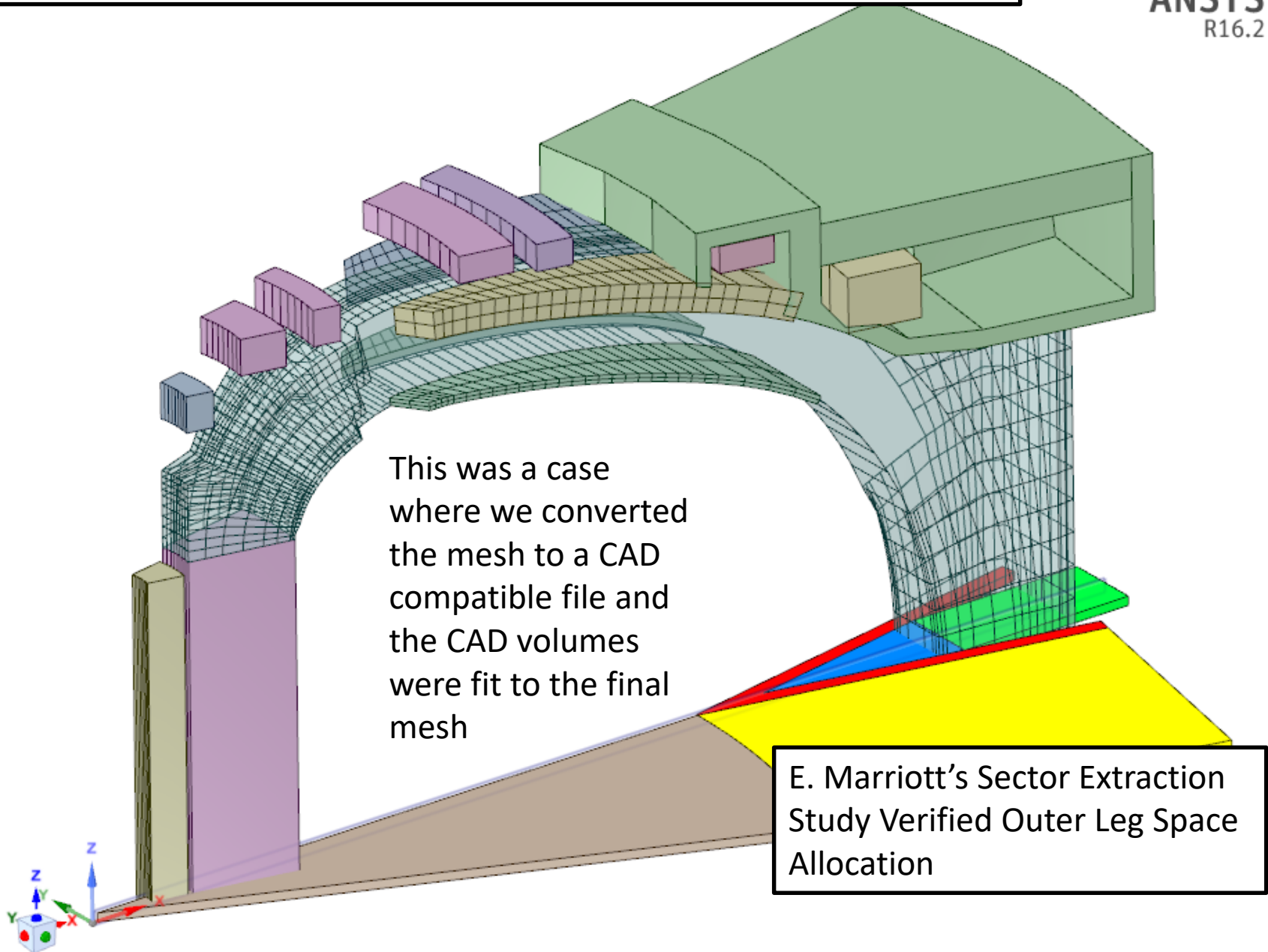


~650 Mpa

DMX =.023895
 SMN =.342E+07
 SMX =.928E+09
 XV =1
 YV =2
 ZV =3
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 *XF =5.07498
 *YF =.283041
 *ZF =-.113E-04
 Z-BUFFER
 .342E+07
 .106E+09
 .209E+09
 .312E+09
 .415E+09
 .517E+09
 .620E+09
 .723E+09
 .826E+09
 .928E+09

A better View of the FEA Model Inserted Into Ed Marriott's Cad Model

ANSYS
R16.2



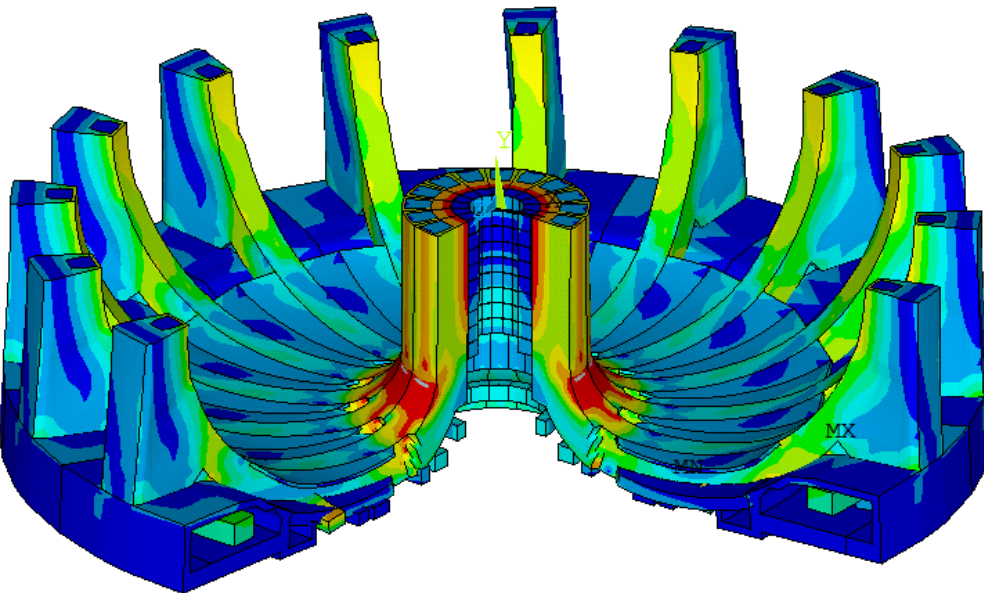
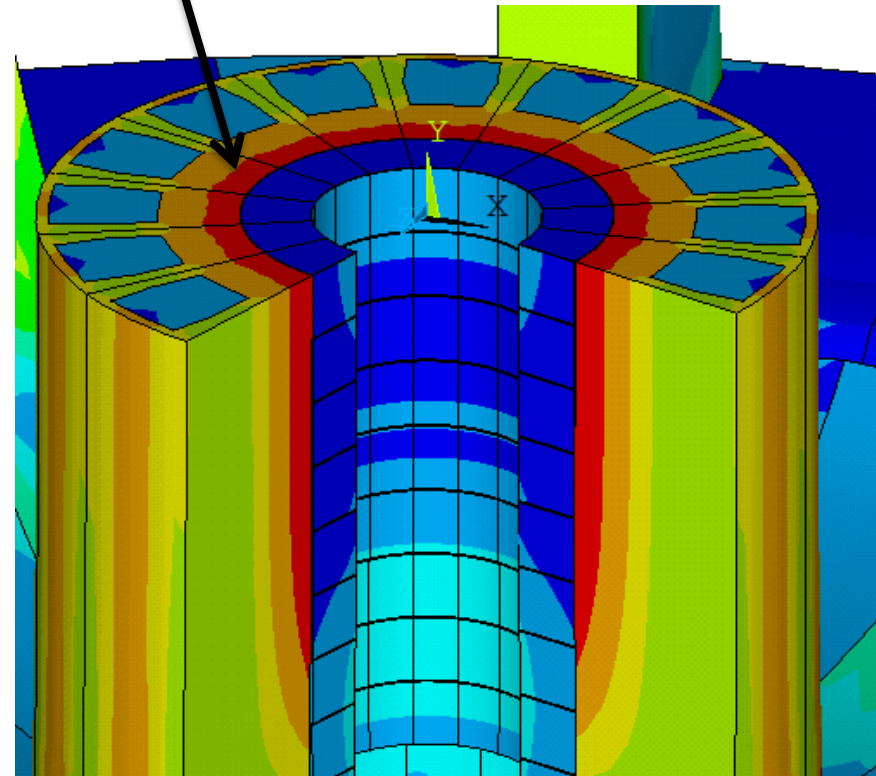
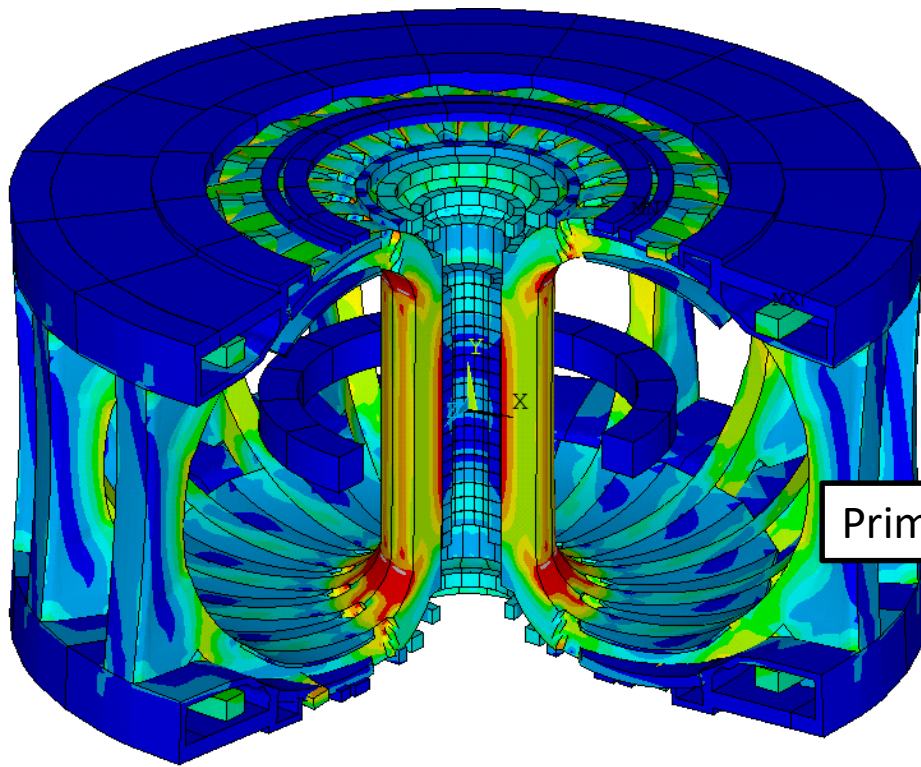
Based on Chucks Aug
28th email 7.5T at Ro
and 1.06 TF Radial
Build. 18 Segment OH,
Added Cap and
External Structure and
"Flared" Outer Leg

SINT

0

.100E+09
.200E+09
.300E+09
.400E+09
.500E+09
.600E+09
.700E+09
.800E+09
.900E+09

Primary Stress is ~800 MPa



1

NODAL SOLUTION

STEP=1

SUB =1

TIME=1

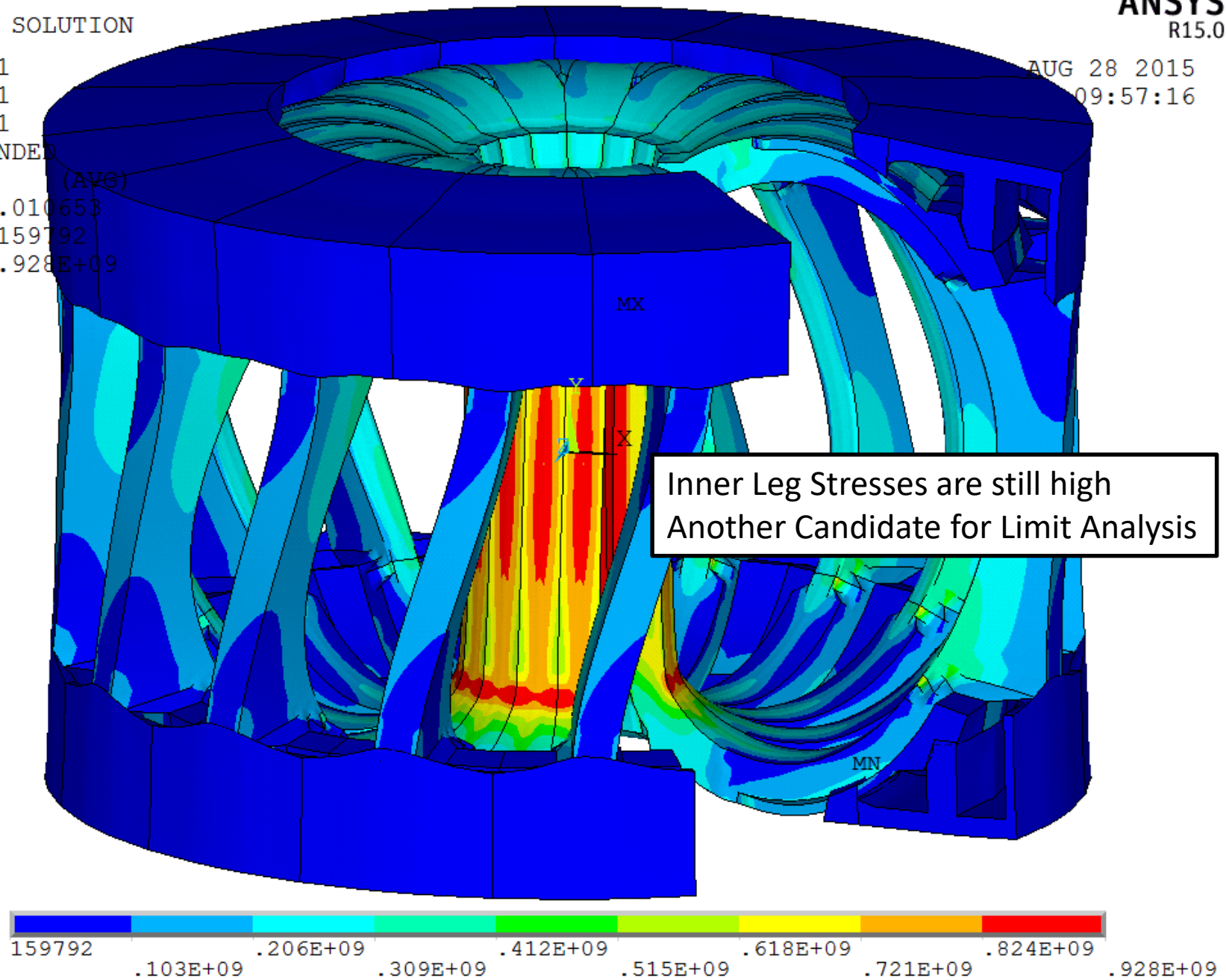
/EXPANDED

SINT (AVG)

DMX =.010653

SMN =159792

SMX =.928E+09



1

NODAL SOLUTION

STEP=1

SUB =1

TIME=1

/EXPANDED

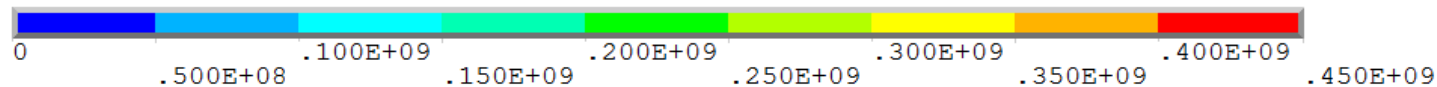
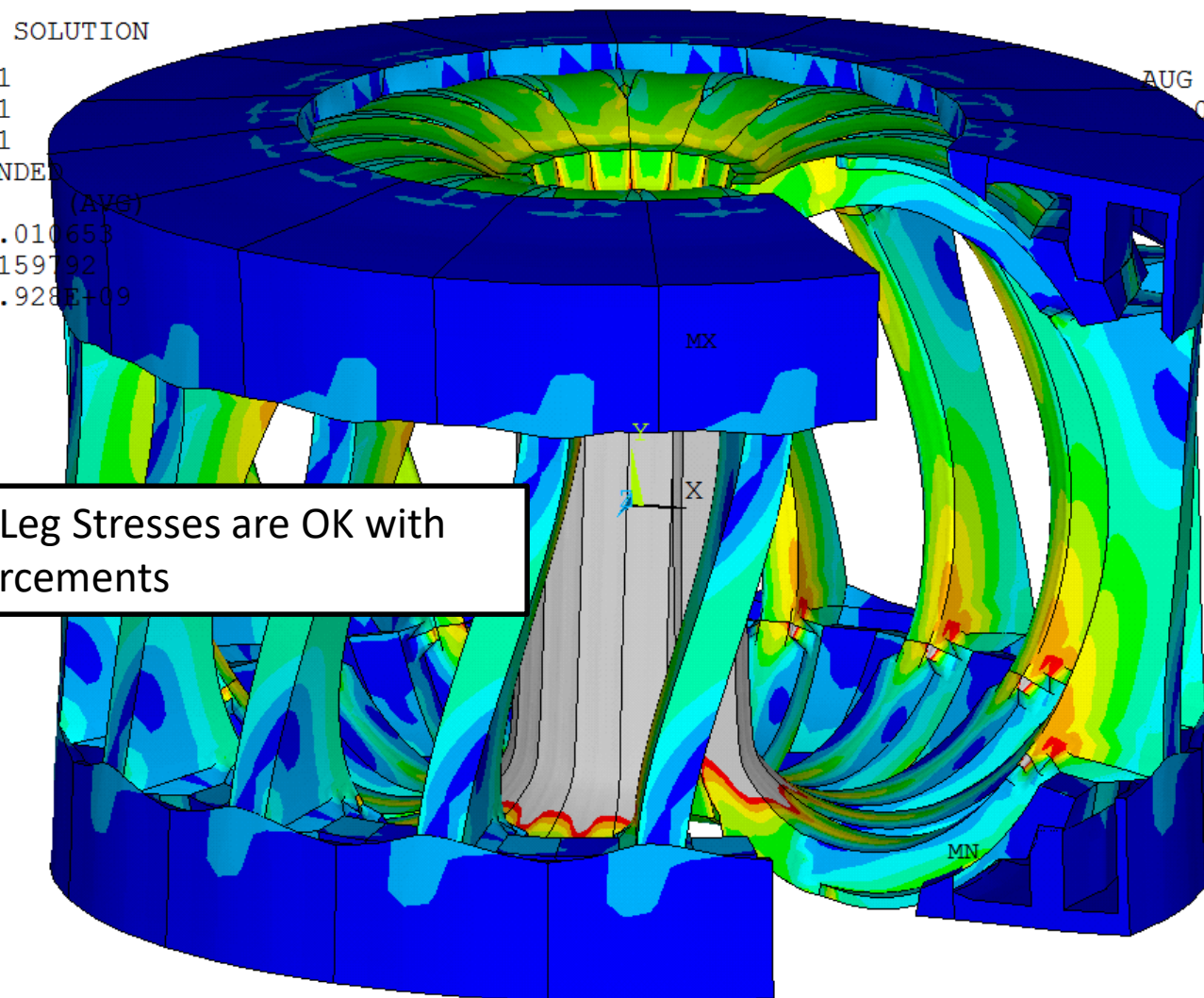
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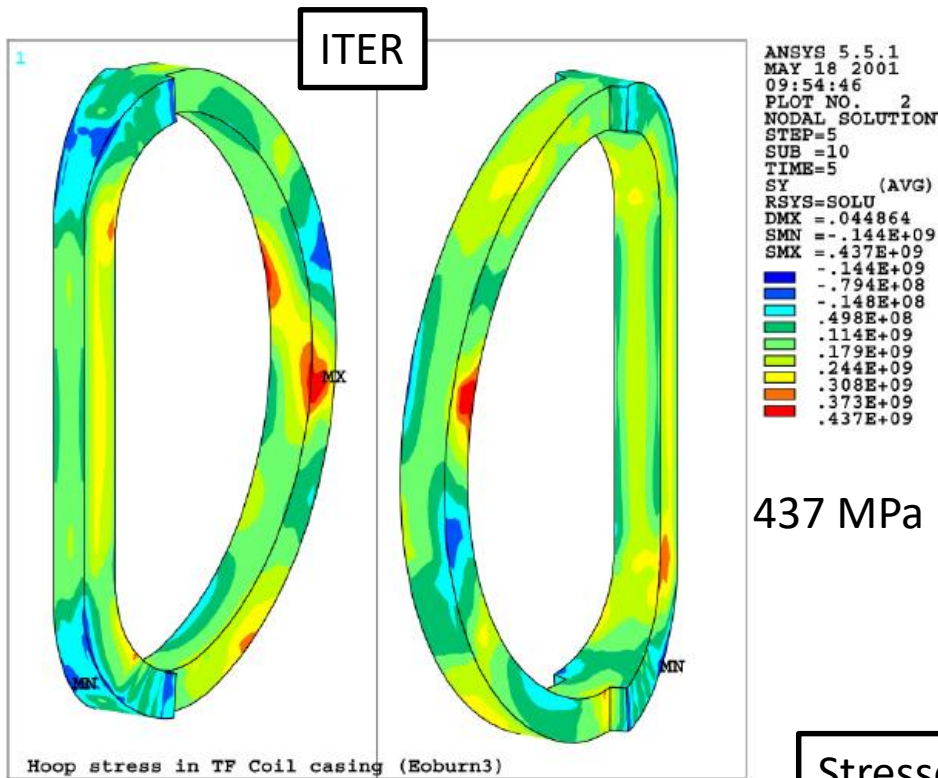
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SMN =159792

SMX =.928E+09

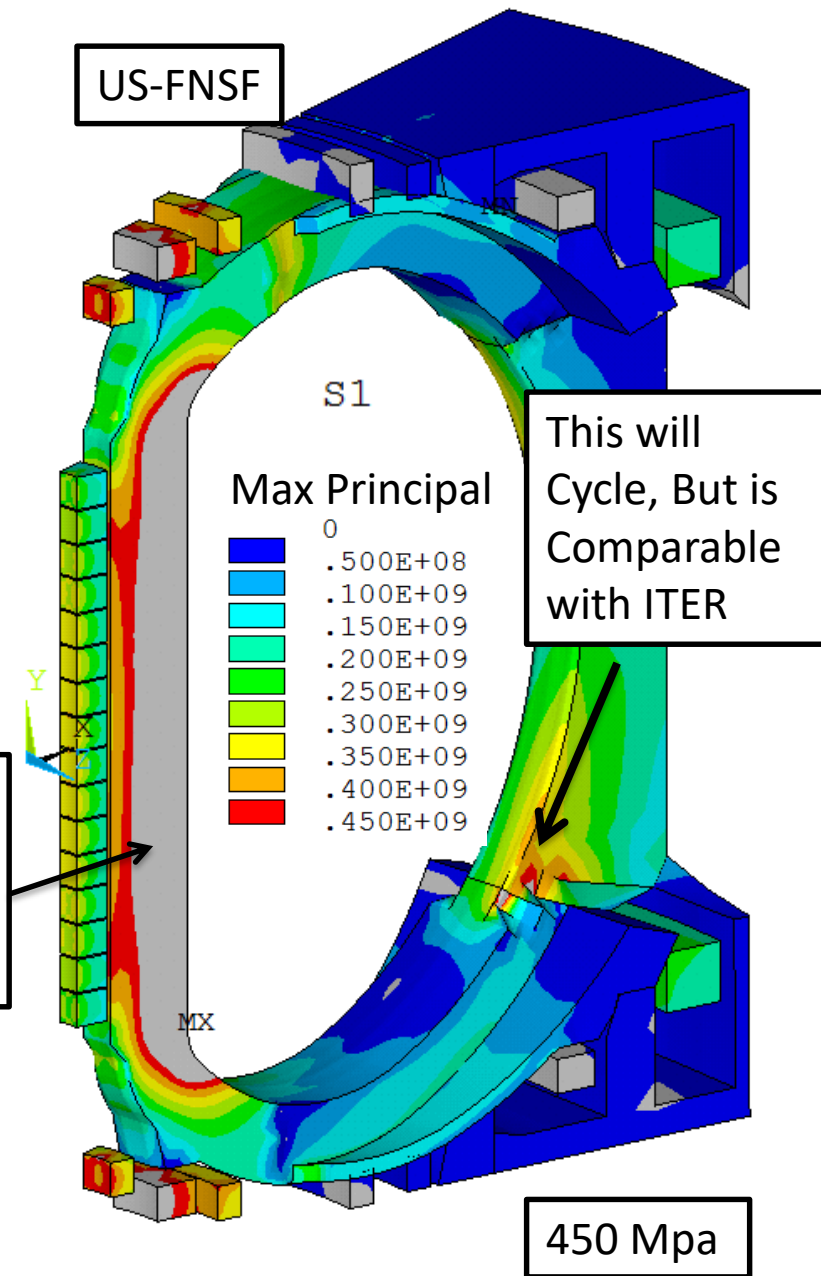
Outer Leg Stresses are OK with
Reinforcements





The Cyclic Tensile Stresses for ITER and US-FNSF are comparable. 437MPa for ITER and 450 MPa for US-FNSF

The US FNSF should not have as high a cyclic requirement as ITER

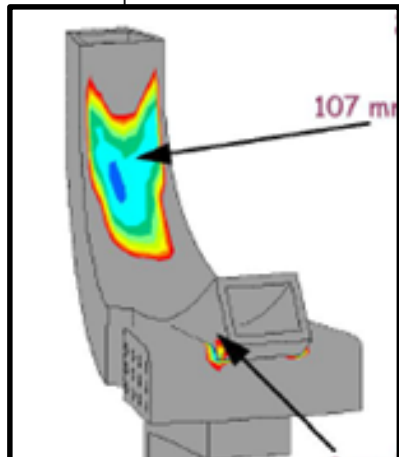


Another Way to Show Cyclic Stress

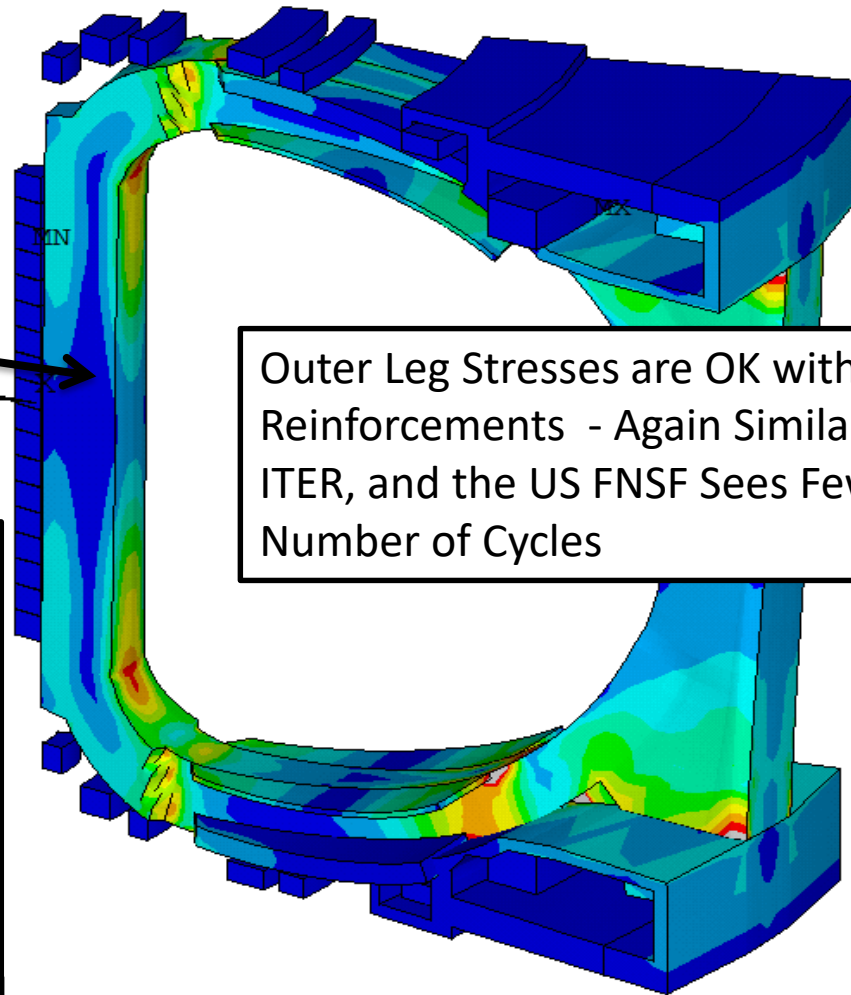
Stress Difference between the TFON and the TFON+PF load case

Inner Leg
Equatorial
Plane
Stresses are
Static

Outer Leg Stresses are OK with
Reinforcements - Again Similar to
ITER, and the US FNSF Sees Fewer
Number of Cycles



Where ITER's Limiting
flaw sizes are.

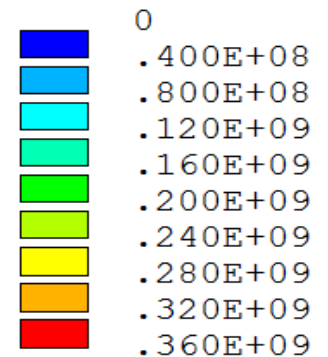


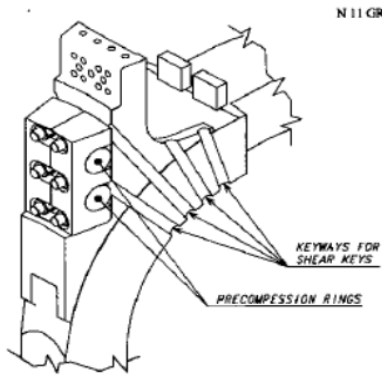
Current Set #2

5.0

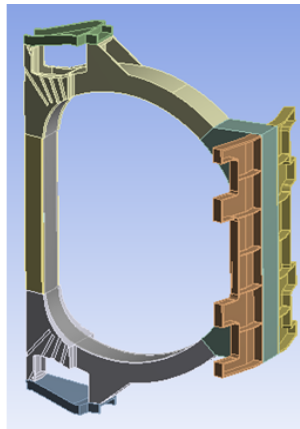
DEC 8 2015
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AVRES=Mat
DMX =.013411
SMN =6.72447
SMX =.116E+10

XV =1
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ZV =2
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XF =6.16876
ZF =-.123E-04
Z-BUFFER



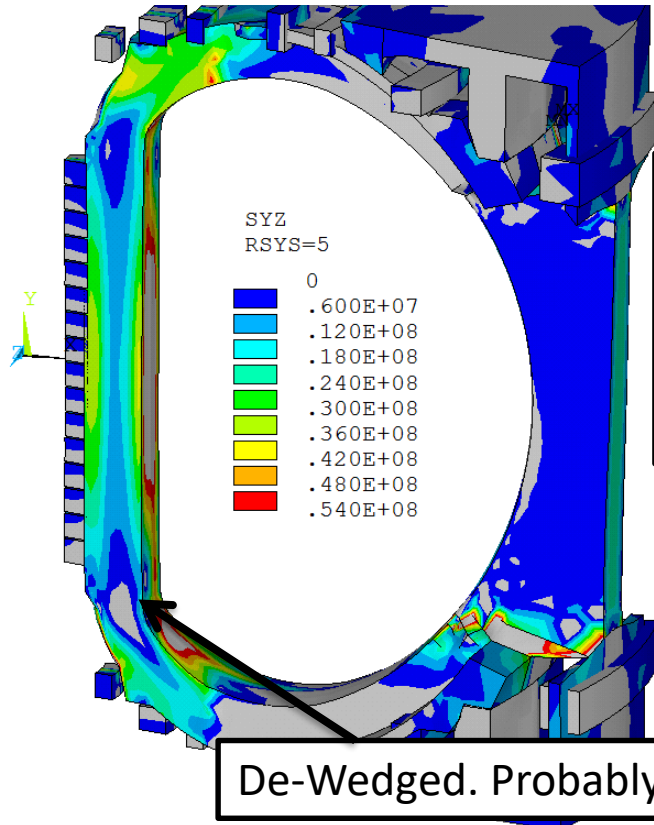


ITER Inner Corner Preload Ring and Shear Keys

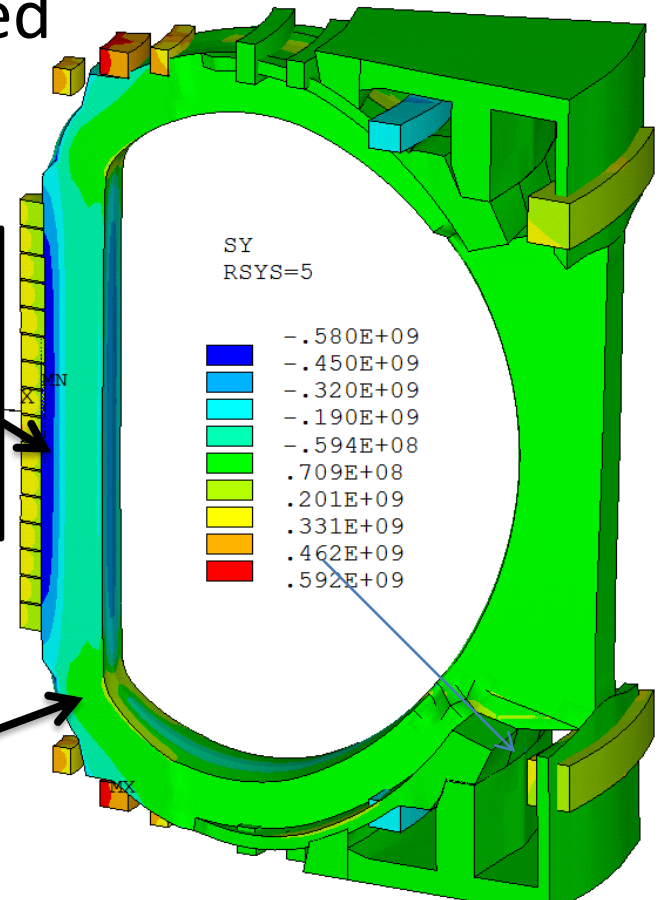


KDEMO Inner Corner Shear Keys

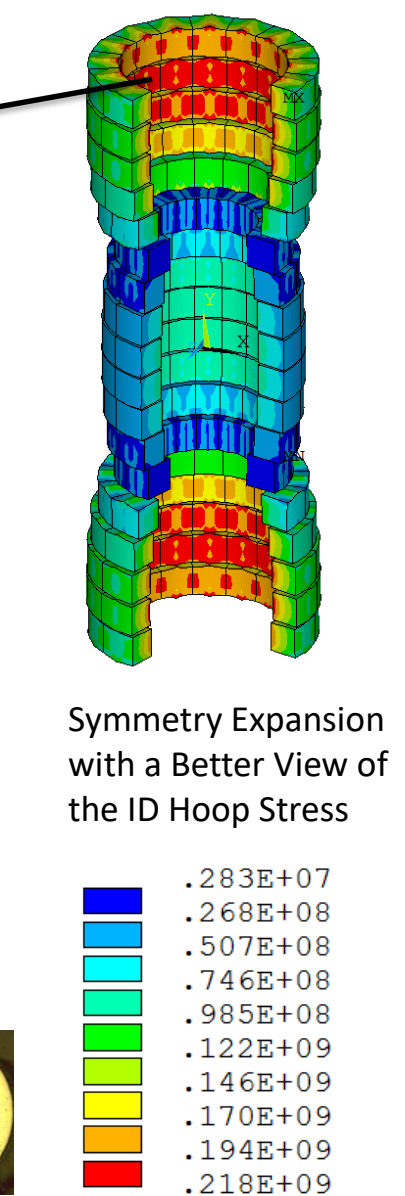
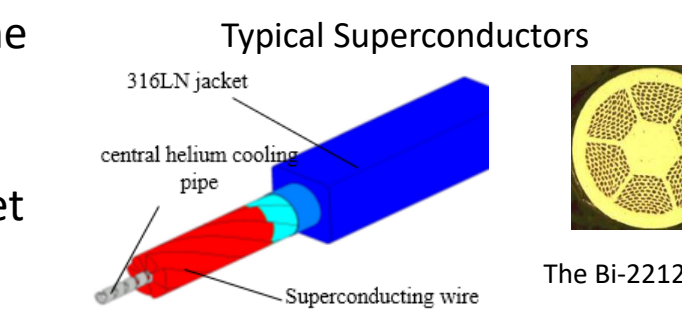
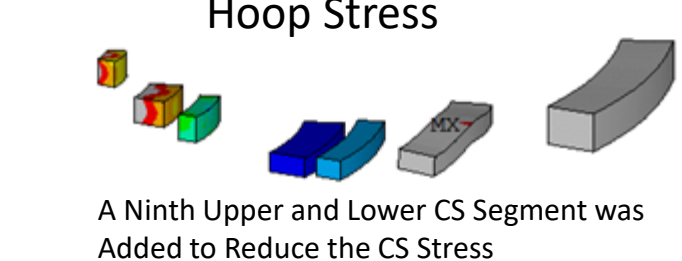
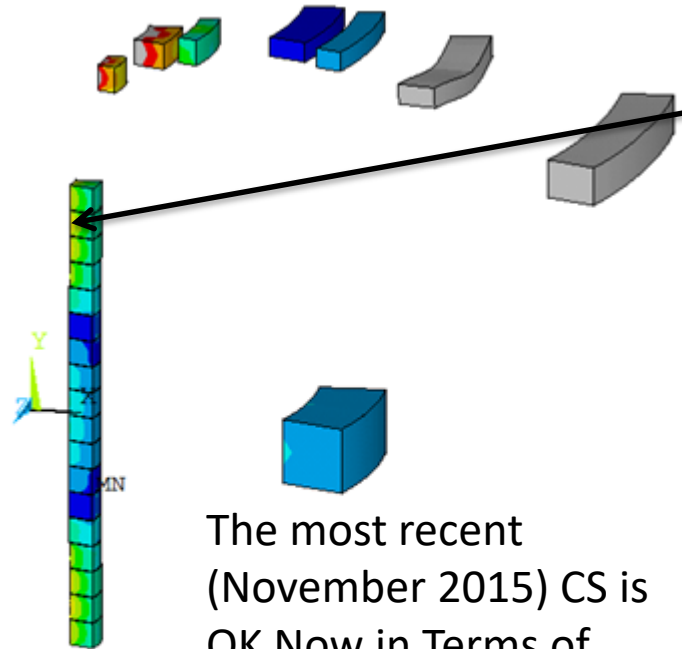
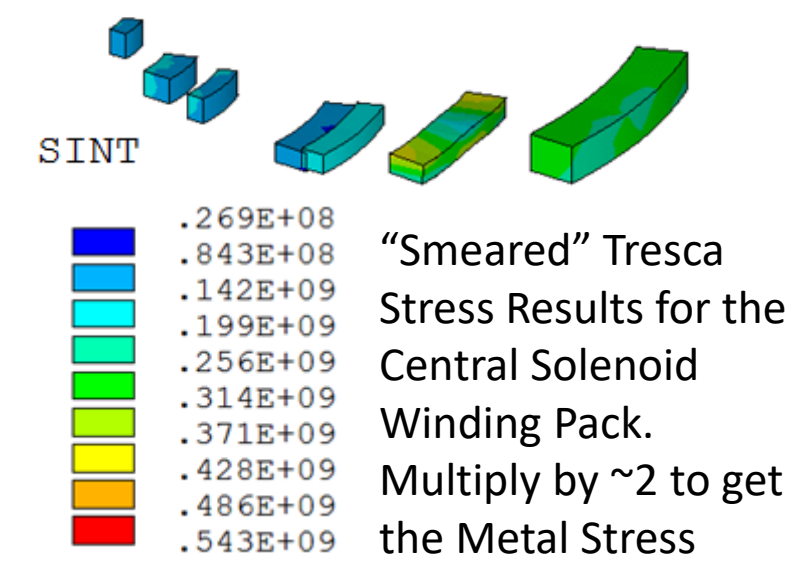
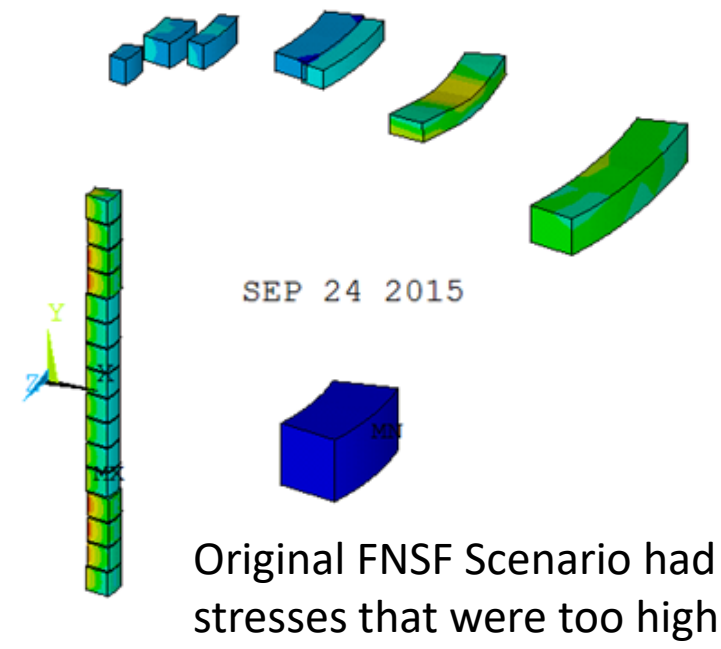
Carrying the Inner Leg
Torsional Shear
FNSF, KDEMO are much
like ITER – Shear Pins are
Needed



Plenty of wedge
pressure at the
nose. Good to
carry 60 to 180
Mpa shear.



Include Full Time Dependent Scenario in Initial Sizing – Not just Equilibria



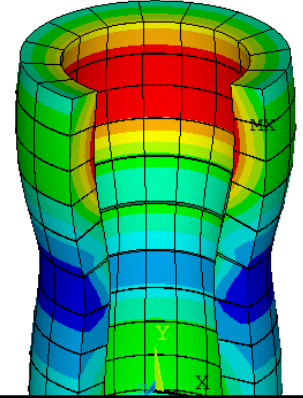
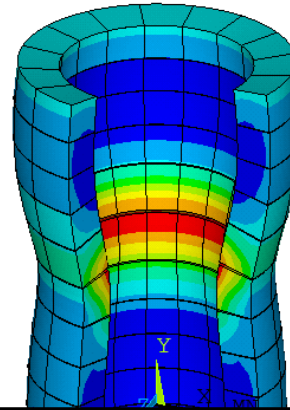
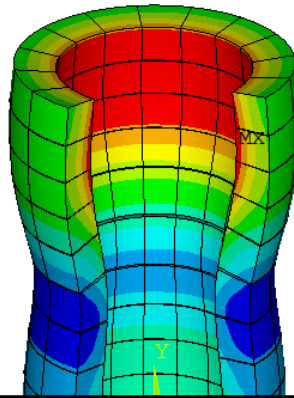
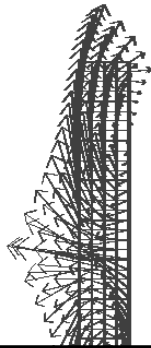
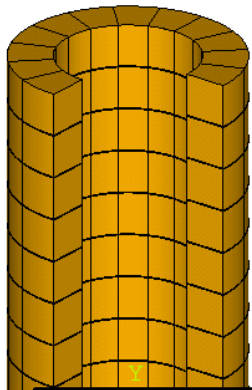
Model

Field

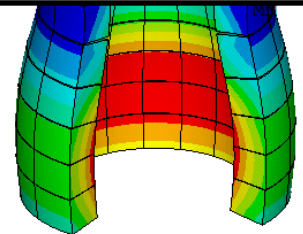
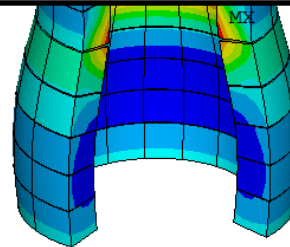
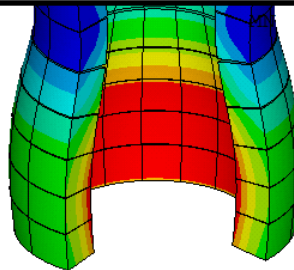
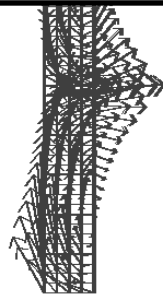
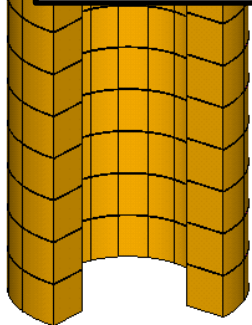
Hoop Stress

Axial Stress

Tresca Stress



To get acceptable CS stresses we had to add a 9th coil, add build and steal space from the TF
To get acceptable TF stresses we had to investigate another coil support scheme

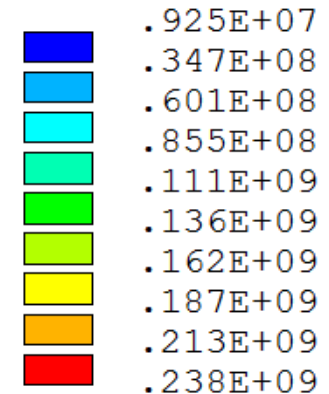
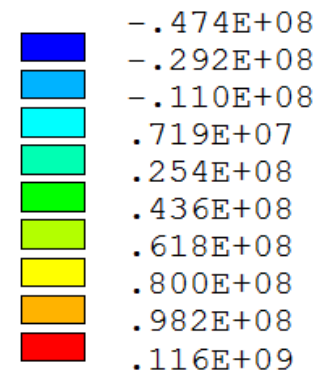
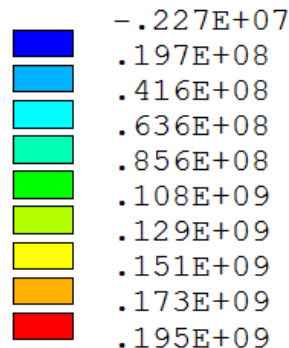


SY (AVG)
RSYS=5

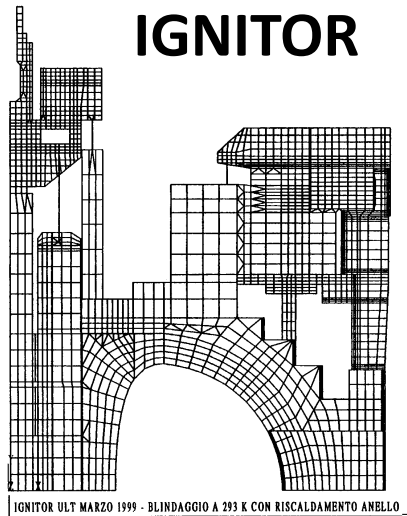
SZ (AVG)
RSYS=5

SINT (AVG)

Model shown with
Symmetry Expansion.
Equatorial Plane
Constraints are Used.
The peak field is 12.65T

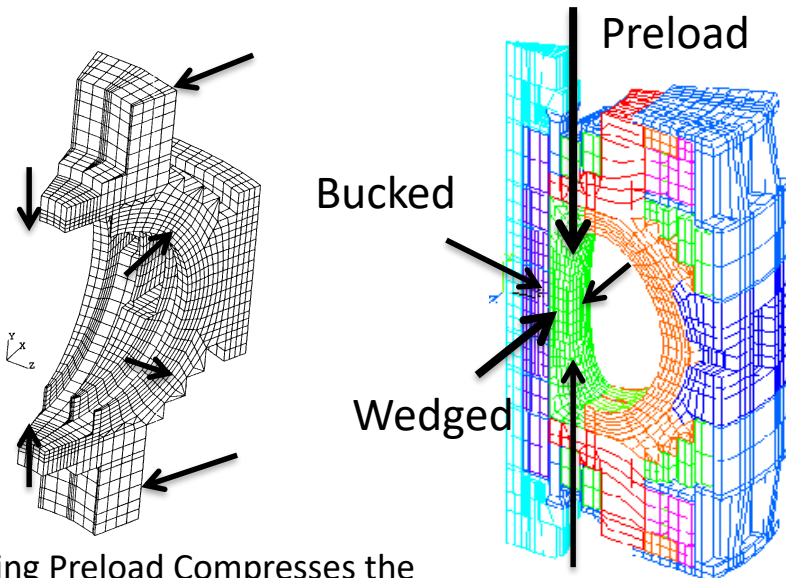
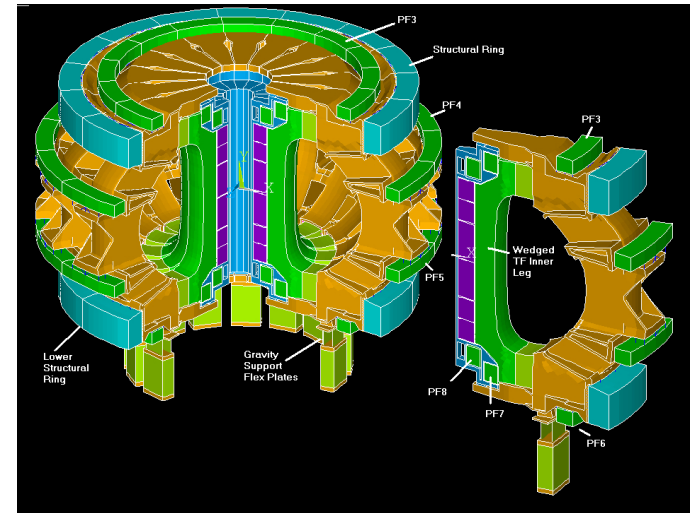


Structural Concepts That Might be Useful for US FNSF and Future HTS



The failure stresses, Tresca and Von Mises are related to the imbalance of stress. If you can approach a hydrostatic state in the inner leg then the tokamak can take larger magnetic pressures.

BUCKED and WEDGED Concepts like IGNITOR take advantage of this.



Ring Preload Compresses the Inner Leg

Concepts with large preload rings like IGNITOR and FIRE seek to take advantage of this – For normal magnets, managing the thermal expansion is difficult and in IGNITOR led to an active preload system. For superconducting magnets thermal expansion during the shot is not a problem. For CICC the He voids make approaching a hydrostatic state impossible. HTS may be better.

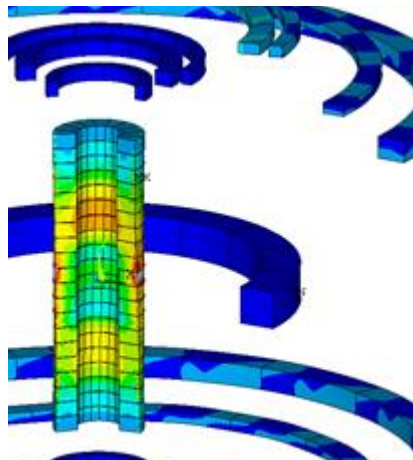
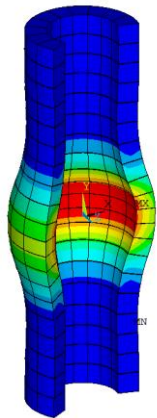
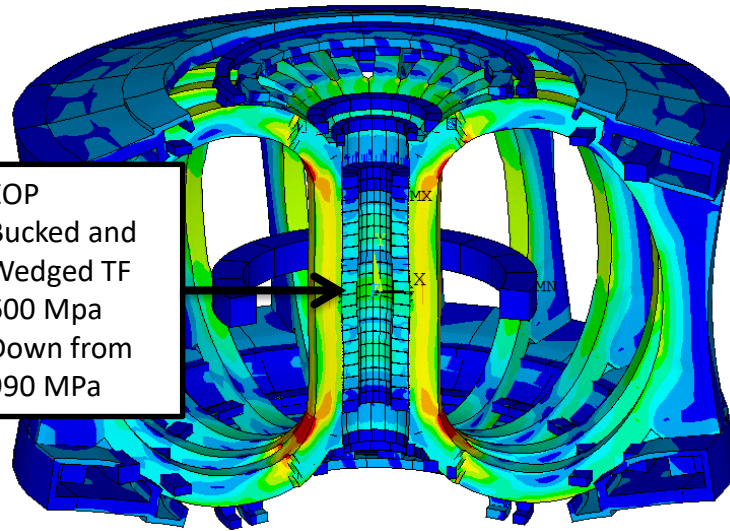
FNSF Bucked & Wedged Solution

P. H. Titus, Jan 10, 2017

Princeton Plasma Physics Laboratory ptitus@pppl.gov

Dec. 2016 – January 2017 – **Buck and Wedge** allows increasing the OH current center radius from .85 to .9. and Increasing the OH build from .4 to .6 m , OH OR= 1.2 m. OH Currents scaled down by .892 (Preserves Volt Seconds). OH Current Densities and Stresses drop by 50%

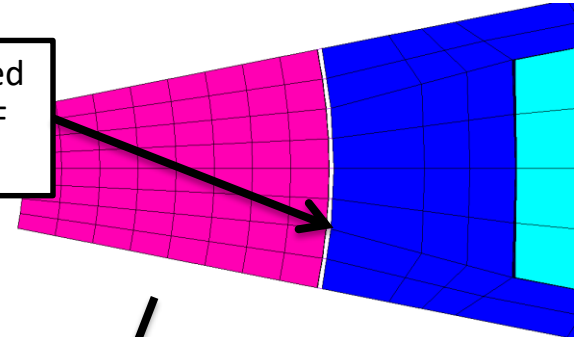
EOP
Bucked and
Wedged TF
600 Mpa
Down from
990 MPa



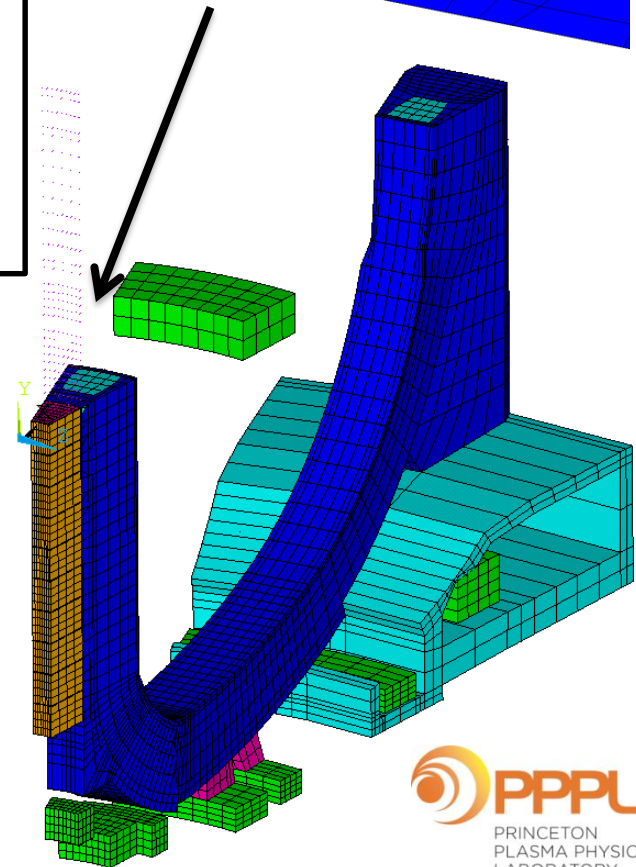
EOP Free Standing ,
Original .4m Build, 1.9GPa

EOP Bucked and Wedged .6m build
, 495 MPa

Gap Elements Added
Between OH and TF
Nose



B & W best
implemented with
Joints, Leads He
Penetrations, and
Preload
mechanisms in the
bore. May be
difficult



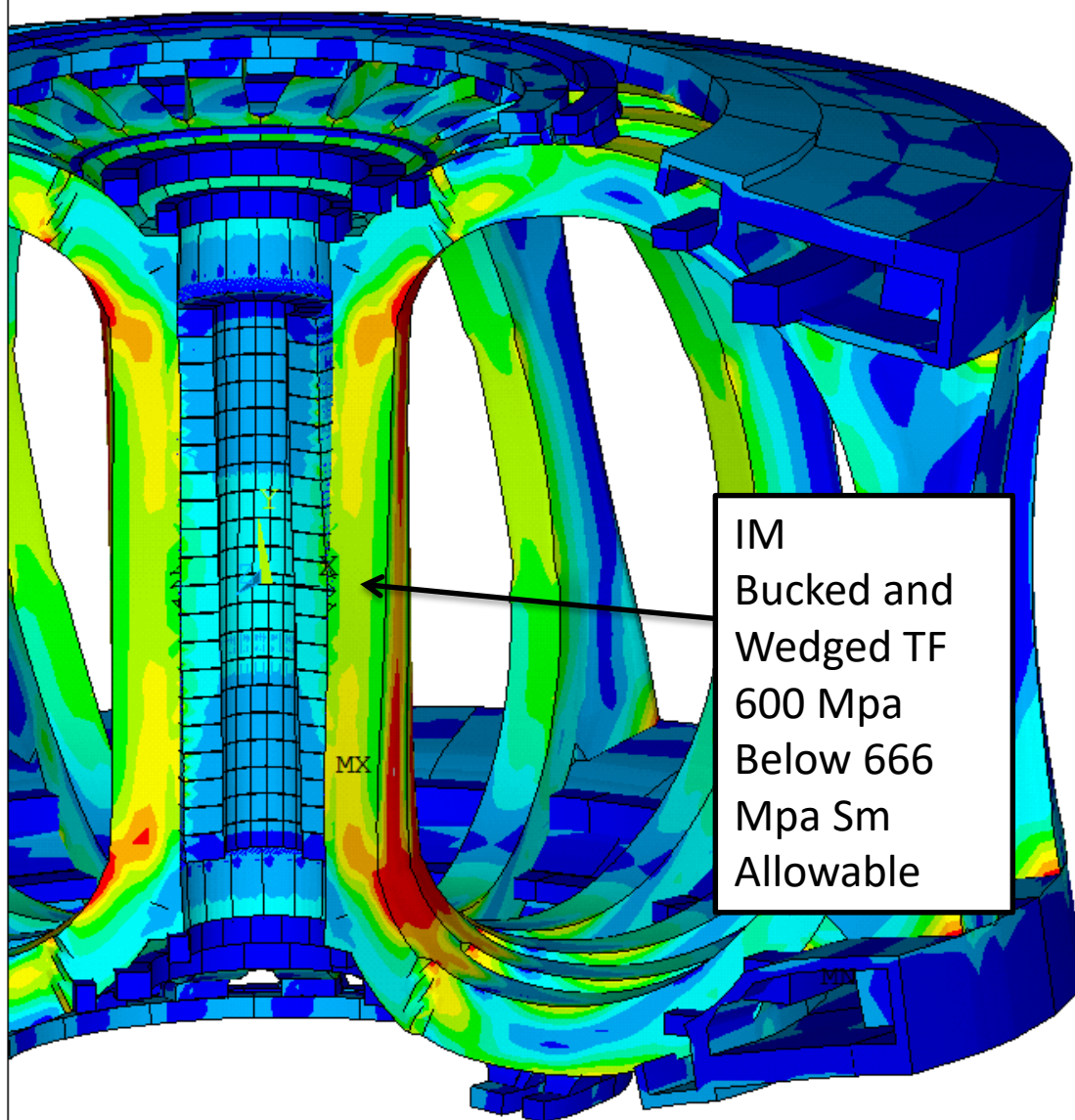
1

ANSYS
R15.0

JAN 4 2017
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 EFACET=1
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 SMN =56213.3
 SMX =.146E+10

XV =.38262
 YV =.200896
 ZV =.9018
 *DIST=9.23498
 *XF =6.13868
 *YF =.006618
 *ZF =.171493
 A-ZS=.904856
 Z-BUFFER

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 .100E+09
 .200E+09
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 .900E+09

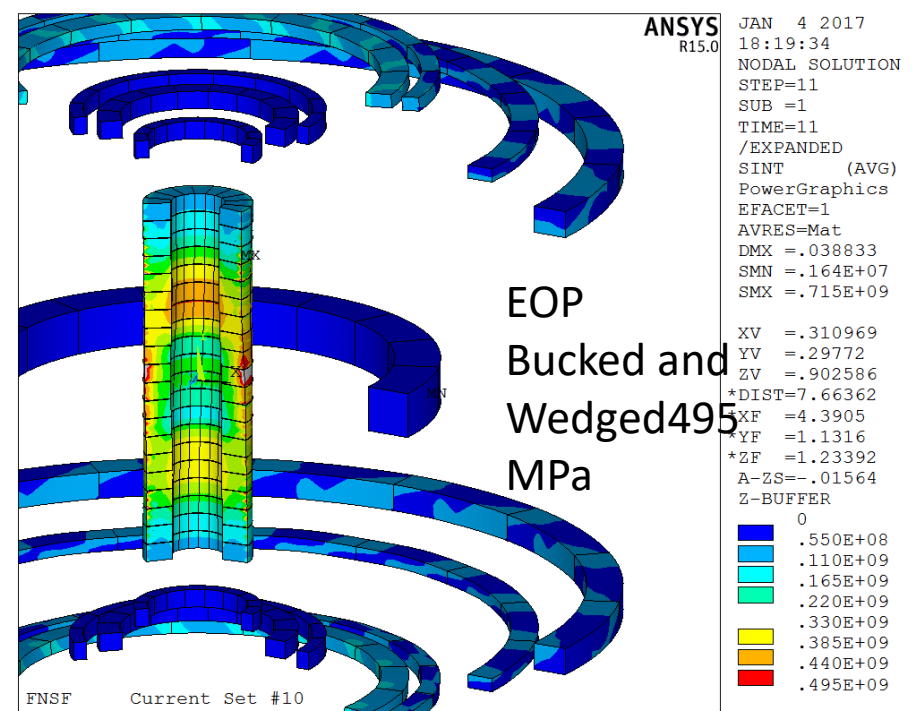
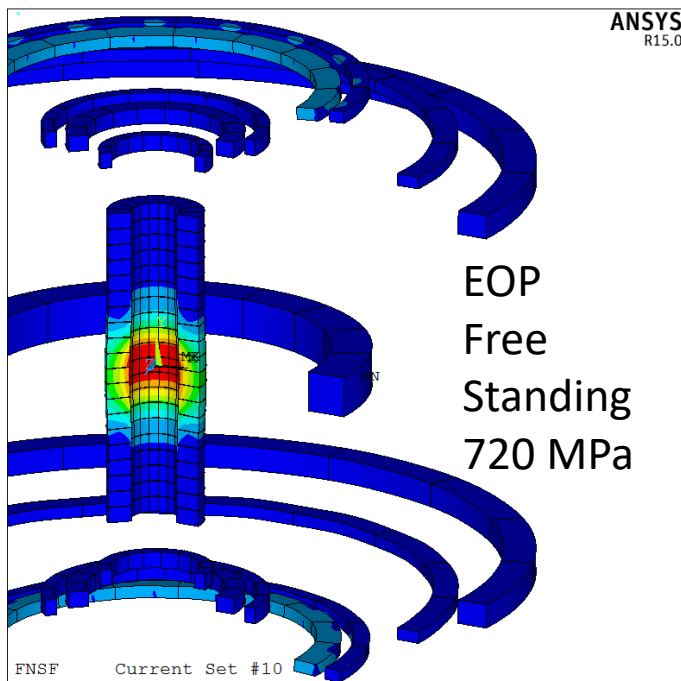
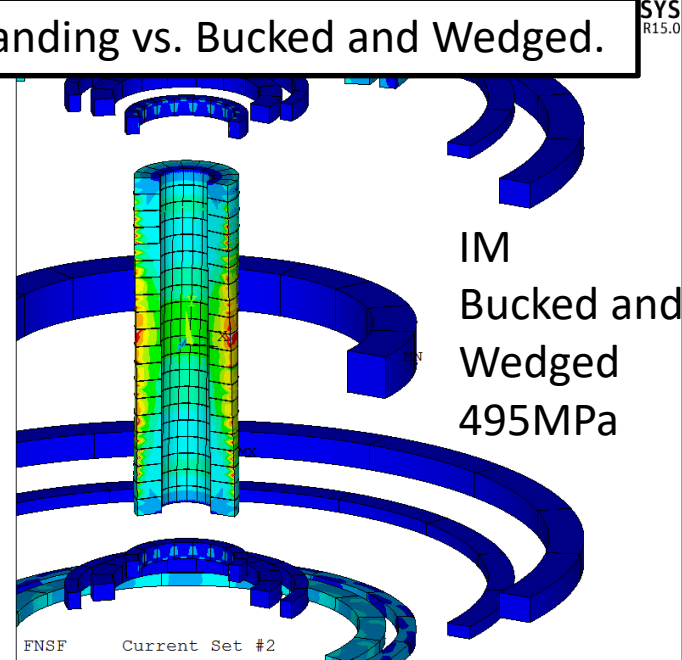
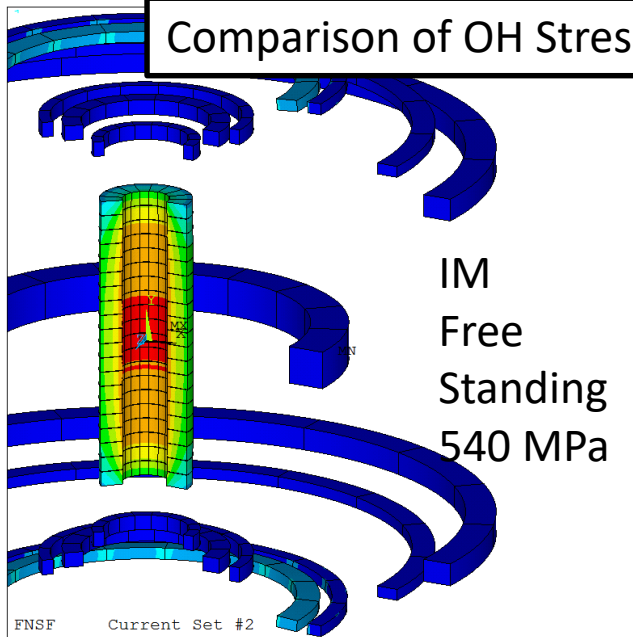


IM TF Stress – Nearly the Same for EOP

FNSF

Current Set #2

Comparison of OH Stresses – Free Standing vs. Bucked and Wedged.



Buck and Wedge vs. Wedge Only

All Results are For OH $r=.9$, $dr=.6m$

Allowable TF Stress =1GPa Peak,666 PM

OH Smeared Static Allowable =330 to 500 MPa Depending on the contribution from the conductor

	Location	IM	EOP	
PF Free Standing Elastic (Smeared)	CS1 ID	540	720	Unacceptable Metal Stress
PF B&W Elastic (Smeared)	CS1 OD	495	495	Closer to Acceptable
TF Free Standing Elastic	Equator Nose	990	990	Acceptable
TF B&W Elastic	Equator Nose	600	600	Acceptable

B & W Improves the stress state of both the OH and TF

OH can be improved by stealing a little more space from the TF

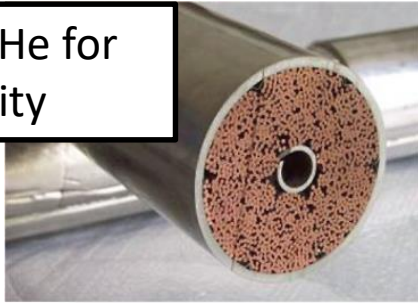
B & W best implemented with Joints, Leads He Penetrations, and preload mechanisms in the bore. May be difficult

US FNSF Conclusion

- With the added radial structure and added structure above and below the horizontal port, the outer leg TF stress is similar to that which was qualified cyclically for ITER (in its inner leg). The FNSF should have less restrictive fatigue requirements. The important conclusion is that RADIAL SERVICING is possible with adequate stress margins for the coil structures.
- Inner leg torsional shear will need features like ITER. Shear keys in the corner and possibly corner tensioned rings.
- Inner Leg Stress is still a bit too high. Some modest reallocation of metal cross sections may still be needed. Improved yield stainless steels are an option. More steel with less space for conductor may be possible with HTS. Limit Analysis was used to qualify K-Demo Stress.

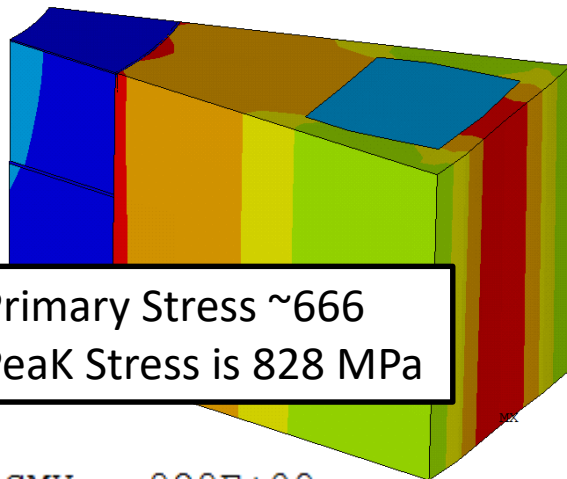
HTS and Solder Filled Nb3Sn Have the Potential of Carrying More Structural Load ?

LTS with He for
AC Stability



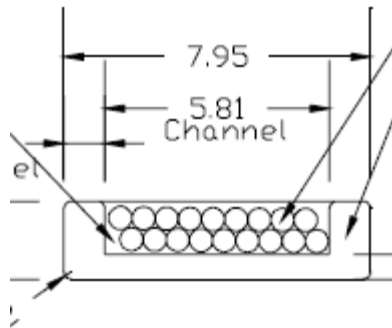
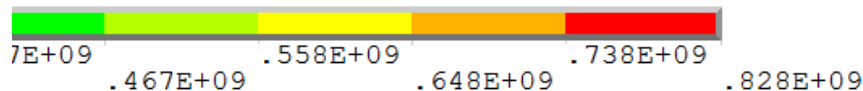
LTS with He for
AC Stability may
not be needed
for low dB/dt in
Demo - Go back
to conduction
cooled react and
wind?

Winding pack modulus
estimated to be 70 Gpa for ITER-
Like CICC , No credit for reacted,
cabled strand



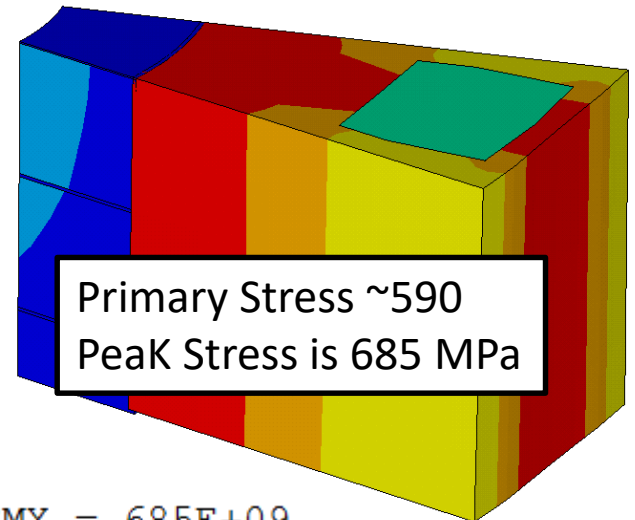
Primary Stress ~666
Peak Stress is 828 MPa

SMX = .828E+09



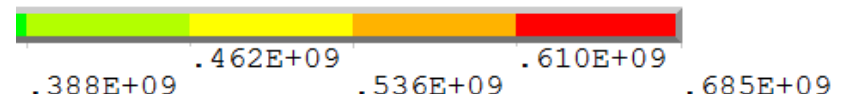
HTS for DC
Operation
Or Very
Long Pulse
Tokamak?

Winding Pack modulus
estimated to be 175 Gpa from
the mixture rule. Essentially all
metal



Primary Stress ~590
Peak Stress is 685 MPa

SMX = .685E+09



Conclusions

- DEMO is approaching a static stress state and requires less emphasis on fatigue.
- Consider retaining S_m , $1.5 S_m$ and $3 * S_m$ Allowables
- Use Limit Analysis to decompose FEA stress into a primary load evaluation.
- Introduce fast fracture checks based on initial NDE of components – use 2 on flaw size – maybe 2 on fracture toughness?
- Radial servicing can be qualified for out-of-plane loads.
- Vertical servicing has similar
- Non-Constant Tension Dee TF Coils can be used to optimize PF performance and global structural requirements
- Remember to size the central column including the central solenoid based on a time dependent scenarios – not just equilibria
- Consider other structural concepts – Bucked or bucked and wedged. Consider preload systems to off load central column stress
- Take advantage of conductors that can contribute to the coil structure –REPCO Tapes, Solder filled Nb3Sn

References

- Peter H. Titus & Ali Zolfaghari (2013) TF Inner Leg Space Allocation for Pilot Plant Design Studies, Fusion Science and Technology, 64:3, 680-686, DOI: 10.13182/FST13-A19171 (Nashville TOFE)
- To link to this article: <https://doi.org/10.13182/FST13-A19171>
- Magnet Structural Design Criteria Part I: Main Structural Components and Welds, Mitchel, Jong, Alekseev
- Fusion Energy Systems Studies (FESS) FNSF Study Year-End Report for 2015 C. E. Kessel¹, J. Blanchard², A. Davis², L. El-guebaly², L. Garrison³, N. Ghoniem⁴, Y. Huang⁴, P. Humrickhouse⁵, Y. Katoh³, A. Khodak¹, S. Malang⁶, N. Morley⁴, M. Rensink⁷, T. Rognlien⁷, A. Rowcliffe³, S. Smolentsev⁴, P. Snyder⁸, M. Tillack⁹, P. Titus¹, L. Waganer⁶, A. Ying⁴, K. Young¹, and Y. Zhai¹ Nucl. Fusion **55** (2015) 053027 (9pp)
- "ITER TF Magnet System Analyses in Faulted Conditions" IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY, VOL. 26, NO. 4, JUNE 2016 Gabriele D'Amico, Luigi Reccia, Alfredo Portone, Cornelis T. J. Jong, and Neil Mitchell
- Design concept of K-DEMO for near-term implementation K. Kim¹, K. Im¹, H.C. Kim¹, S. Oh¹, J.S. Park¹, S. Kwon¹, Y.S. Lee¹, J.H. Yeom¹, C. Lee¹, G-S. Lee¹, G. Neilson², C. Kessel², T. Brown², P. Titus², D. Mikkelsen² and Y. Zhai²
- Magnet Design Considerations for Fusion Nuclear ... - IEEE Xplore ieeexplore.ieee.org/iel7/77/6353170/07415959.pdf Y. Zhai, C. Kessel, L. El-Guebaly and P. Titus
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Back-Up Slides

Orthotropic “smeared” Material Properties of the TFWP Used in 3D Global Non-linear Model (ITER_D_2MVZNX)

E_x	60.7GPa
E_y	100.GPa
E_z	48.9GPa
G_{xy}	27.2GPa
G_{yz}	22.7GPa
G_{xz}	6.44GPa
ν_{xy}	0.239
ν_{yz}	0.243
ν_{zx}	0.159
α_x (for 293K to 4K)	0.304%
α_y (for 293K to 4K)	0.299%
α_z (for 293K to 4K)	0.318%

		R0	Bo	Peak TF Field (T)	Num TF	Cur per TF Coil MAT	Winding Pack Area, m^2	Winding Pack Current Density, MA/m^2	Inner Leg TF Stress, Primary Membrane Tresca Mpa	
FNSF		4.8	7.5	17.57	16	11.25	0.48	23.4375	800	
ITER		6.2	5.3	10.948	18	9.128	0.5344	17.0808383	705	
HTST	9/15/2014			13.165		2.24	0.0303659	73.7669071		
ST 3 Meter		3	7	16.655	16	6.563	0.17518	37.4643224		
ST 3 Meter		3	7	17.023	10	10.5	0.2534	41.437648		
KDEMO		7	7	16.87	16	15.313	1	15.313	850	
CFETR		5.7	5	9.577	16	8.9065	0.522291	17.0527541	540	
CFETR HTS							1	0		
								#DIV/0!		

Base cable: 40 tapes, 4 mm width, 0.1 mm YBCO Tape Ic (4.2K, 20T) = 170 A

MIT TSTC Conductor – Achieved at NHMFL

6 kA TSTC $0.01 \times 0.01 \text{ m} = 1\text{e-}4 \text{ m}^2$



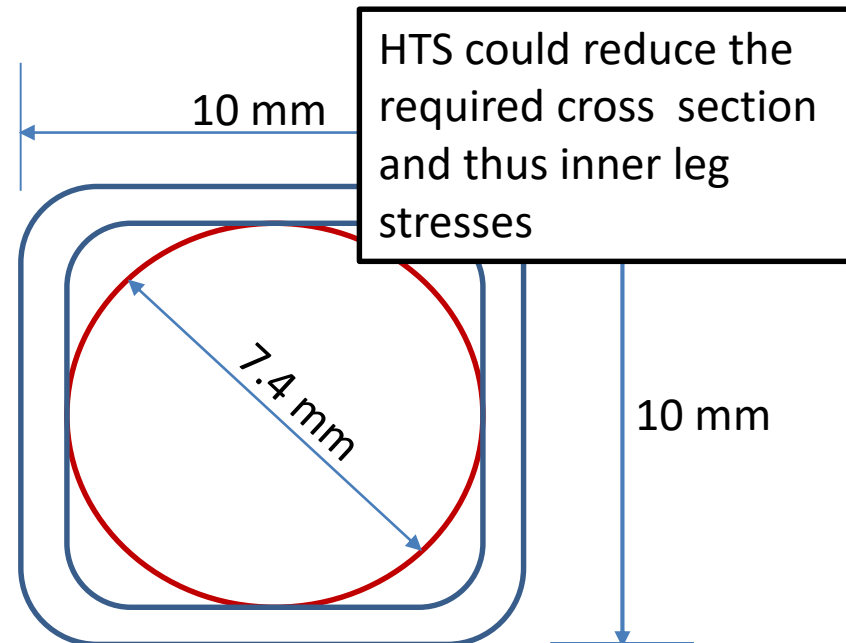
1000 turns $\rightarrow A = 0.1 \text{ m}^2$

Overall $J_e = 60 \text{ A/mm}^2$

Achieved	I_c	Overall J_e
TSTC tested at NHMFL STTW coil, Soldered with braided copper	6.0 kA (B=17 T, T=4.2 K)	117 A/mm ² (B=17 T)

$$J_e(4 \text{ mm} \times 4 \text{ mm}) = 375 \text{ A/mm}^2$$

$$J_e(5.7 \text{ mm Dia.}) = 239 \text{ A/mm}^2$$



....Just to Document the TF Radii and Centers and PF r,z,dr,dz Used

FNSF

TF and PF Structural Analysis Results

Peter H. Titus, Princeton Plasma Physics Laboratory

20151013 14:36:45

FNSF has 16 TF Coils, with 1 turns per coil

FNSF has a major radius of 4.8 m with a toroidal field of 7.5 at Ro

FNSF has a minor radius of 1.2 m

Case Radius set to 3 Case Width set to .8035

OIS Radius set to 3 Case Width set to 3

Section filename=wba4 divx,divy= 2 2

Case Radius set to 2.8 Case Width set to .8035

Nose radius set to 1.1

FNSF Path has 7 Points in the TF Path

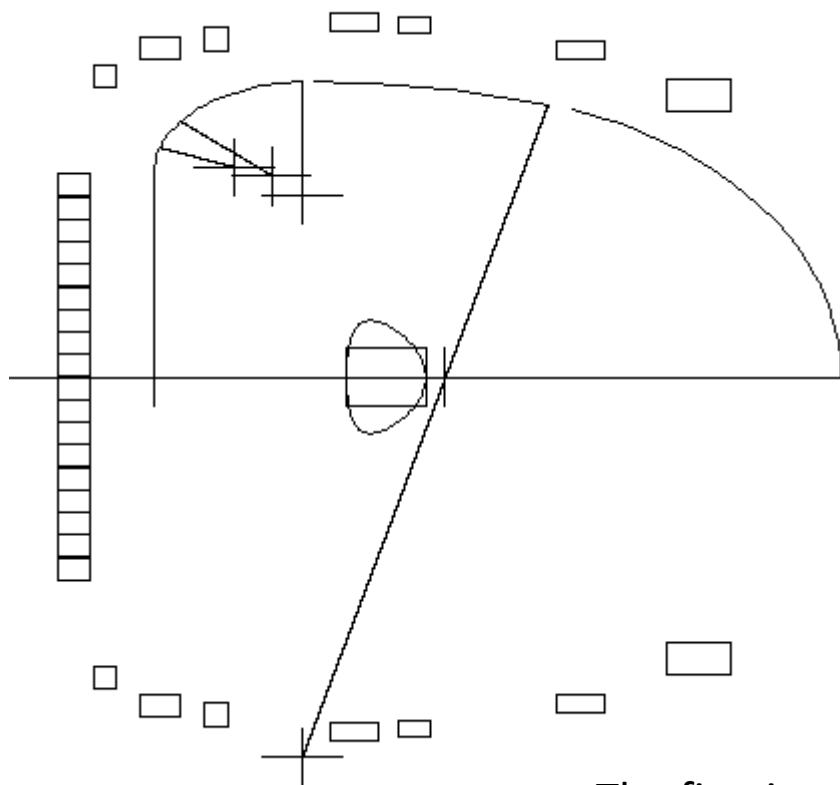
FNSF has 33 Poloidal Field (PF) Coils

FNSF has 2 Poloidal Field (PF) Currents in the Scenario

Scenario 2 is being analyzed

Each TF sector is 22.5 degrees

The current per TF coil is: 11250358. amps



!Pr Pz Pang FNSF Path Specs
PATH

7

s,1.8833, 0, 2, 0

t,0, 3.7309231, 0, 20

r,2.8833, 3.7309231, 20, 5

r,3.3531463, 3.559913, 20, 10

r,3.7361685, 3.2385192, 50, 20

r,3.7361685, -6.7614808, 15, 20

r,5.5479018, -4.8262461e-10, 75, 20

The first inner
corner radius
was chosen as
3.7309231-
2.8833=.8476m

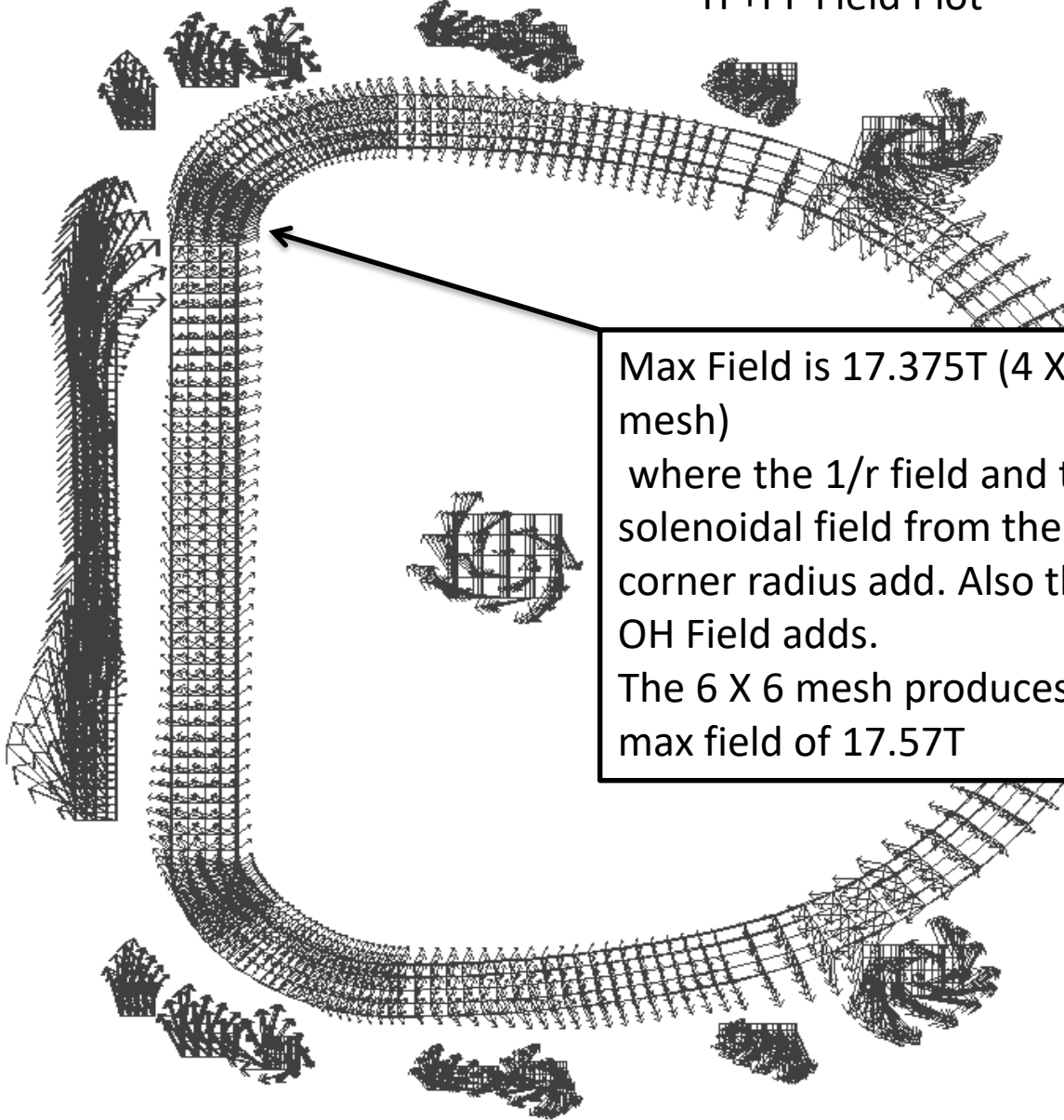
PF R,Z,ND,DZ,NX,NY

1, .85, .2, .4, .39, 4, 4
2, .85, .6, .4, .39, 4, 4
3, .85, 1, .4, .39, 4, 4
4, .85, 1.4, .4, .39, 4, 4
5, .85, 1.8, .4, .39, 4, 4
6, .85, 2.2, .4, .39, 4, 4
7, .85, 2.6, .4, .39, 4, 4
8, .85, 3, .4, .39, 4, 4
9, .85, 3.4, .4, .39, 4, 4
10, 1.25, 5.35, .3, .4, 4, 4
11, 1.95, 5.85, .5, .4, 4, 4
12, 2.65, 6, .3, .4, 4, 4
13, 4.4, 6.3, .6, .3, 4, 4
14, 5.15, 6.25, .4, .3, 4, 4
15, 7.25, 5.8, .6, .3, 4, 4
16, 8.75, 5, .8, .6, 4, 4

PF, Cur 1, Cur2

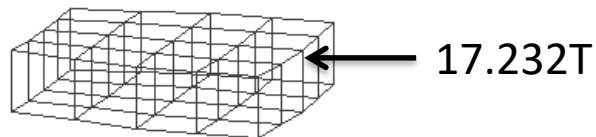
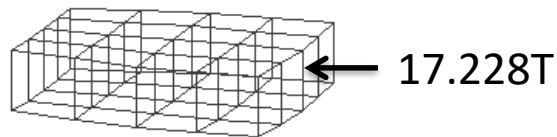
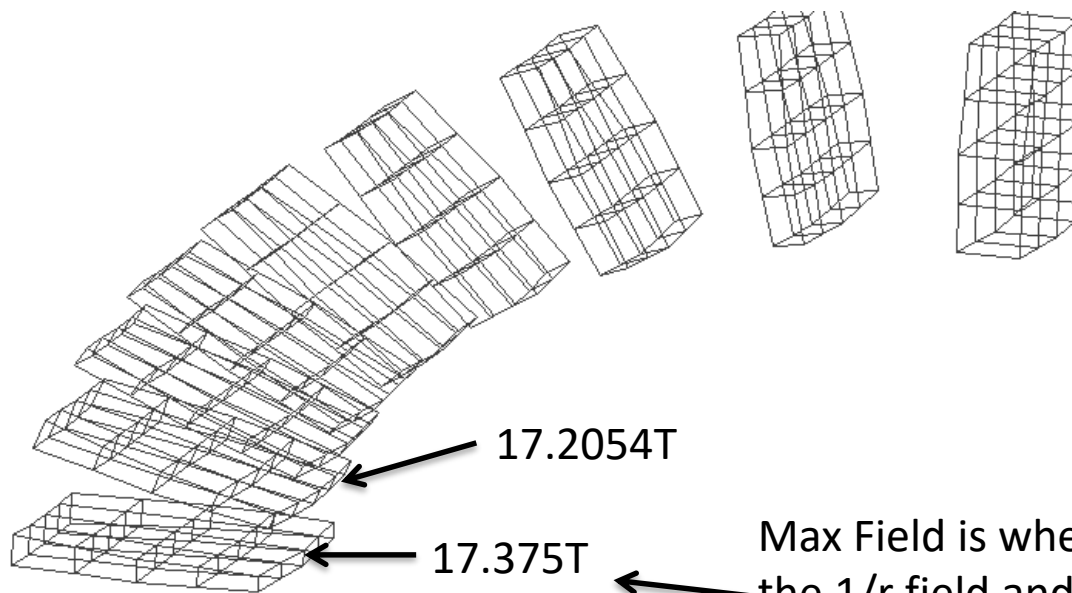
1, 0, -4.42, 0, 0, 0, 0, 0
2, 0, -4.42, 0, 0, 0, 0, 0
3, 0, -4.42, 0, 0, 0, 0, 0
4, 0, -4.42, 0, 0, 0, 0, 0
5, 0, 4.67, 0, 0, 0, 0, 0
6, 0, 4.67, 0, 0, 0, 0, 0
7, 0, 4.67, 0, 0, 0, 0, 0
8, 0, 4.67, 0, 0, 0, 0, 0
9, 0, 4.67, 0, 0, 0, 0, 0
10, 0, 4.04, 0, 0, 0, 0, 0
11, 0, 5.98, 0, 0, 0, 0, 0
12, 0, 6.91, 0, 0, 0, 0, 0
13, 0, 1.33, 0, 0, 0, 0, 0
14, 0, 2.32, 0, 0, 0, 0, 0
15, 0, 7.23, 0, 0, 0, 0, 0
16, 0, -14.36, 0, 0, 0, 0, 0

TF+PF Field Plot



Max Field is 17.375T (4 X 4 mesh)
where the $1/r$ field and the solenoidal field from the corner radius add. Also the OH Field adds.
The 6 X 6 mesh produces a max field of 17.57T

One drawback of the TF Shape is the Peak Field in the Corner – Greater than the quoted 16 T



Max Field is where
the 1/r field and
the solenoidal
field from the
corner radius add.
The 6 X 6 mesh
produces a max
field of 17.57T

```
getn
Node Number
Dir      Cart
x      2.241210
y      3.964630
z      -2.0176500E-01
fx      -243306.7
fy      -7717.539
fz      85473.95
t      0.0000000E+00
bt,bx,by,bz      1
]
Node Number
Dir      Cart
x      2.201660
y      3.778590
z      -1.9363300E-01
fx      -285239.5
fy      2910.226
fz      19956.48
t      0.0000000E+00
bt,bx,by,bz      1
]
Node Number
Dir      Cart
x      2.200000
y      2.984740
z      -1.5893301E-01
fx      -991340.3
fy      45027.63
fz      -1.7983946E-01
t      0.0000000E+00
bt,bx,by,bz      1
]
```

Field is slightly above 17T because of local geometry effects. Reduce ripple, and reduce sharp TF inner corner radius